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# A look into the user experience of commercial indoor farmers

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# En undersökning av användarupplevelsen hos kommersiella inomhusodlare

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## *Sammanfattning*

Med en ökande befolkning ökar även det globala matbehovet. Världsbefolkningen förväntas öka till 8.5 miljarder människor till år 2050. I har jordbruksindustrin behövt förstärka sina resurser för att öka livsmedelsproduktionen. Ett sätt det sker på är genom utveckling av inomhusjordbruk. Odling av mat inomhus kan leda till förbättrad livsmedelssäkerhet, vilket skulle vara en positiv utveckling i det långa loppet.

Inomhusodlingen utvecklas under åren och syftet med detta projekt är att utforska hur inomhusjordbruket påverkar människorna som odlar dessa grödor, det vill säga inomhusodlarna. Projektets fokus är att förstå de utmaningar som en inomhusodlare står inför genom hela odlingsprocessen. För att genomföra användarstudierna används en människocentrerad designprocess (eng. "human-centered design process"). Användarstudierna använder sig av en triangulering av metoder som inkluderar djup- intervjuer, enkäter, samt andra metoder för att få en djupare förståelse för odlarna och deras perspektiv. Tre s.k. kundresor "Journey maps" skapas, där kundresan används för att identifiera smärtpunkterna under odlingsprocessen. Insikterna från den kvalitativa analysen har kvantifieras för att fastställa hur odlarnas dagliga aktiviteter ser ut, samt vad deras utmaningar är under odlingsprocessen. Baserat på dessa insikter skapas två koncept för att erbjuda lösningar för att förbättra odlarens upplevelse. Användartesterna visar att lösningarna var önskvärda men de bör validieras kvantitativt för att säkerställa deras giltighet. Även forskningsfrågorna och eventuellt framtida arbete diskuteras.

**Nyckelord:** Människocentrerad designprocess, **I**nomhusodlare.

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## *Abstract*

As the population increases the demand for food also increases. It is estimated that the population will reach over 8.5 billion by the year 2050. The farming industry is pooling resources to increase food production by developing indoor farming. This practice of growing food indoors can attend to the food security issue which is a positive development in the long run.

Indoor farming has been developing over the years and the purpose of the project will be an exploration of how indoor farming is affecting the people who grow these crops, i.e., the indoor farmers. The focus of the project is to understand what are the challenges and pain points of an indoor farmer, in the whole growing process. The human-centered design process is opted in this project to conduct the user studies. The user studies use methodological triangulation which includes in-depth interviews, surveys, and additional methods to gain a deeper understanding of the farmers' perspective. Three Journey maps are created out of which one journey is used to identify the pain points in the growing process. The Insights generated from the qualitative analysis are quantified to determine what the farmers' daily activities look like, and their challenges during the growing process. On the basis of these insights, two concepts are created to offer solutions to improve the farmer's experience. User tests determine that the solutions were desirable but should be quantified to check the validity. The research questions and future work are discussed further.


**Keyword:** Human-centered design, Indoor farmer.

# FOREWORD

I would like to start by thanking Green AB for giving me this opportunity to conduct a master thesis. Many thanks to my colleagues at Green Ab for creating a fun work environment.

I would like to thank my academic Supervisor Sara Ilstedt at KTH Royal Institute of technology for being so kind to me and guiding me through the thesis. Special thanks to my industrial supervisor for supporting and encouraging me in every step of my thesis.

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Finally, I would like to thank my family and friends for their love and support.

Sanketh Yeli

Stockholm, February 2022

# Nomenclature

## *Abbreviations*

| | |
|---|---|
| *HCD* | Human-Centered Design |
| *CEA* | Controlled Environment Agriculture |
| *DWC* | Deep Water Culture |
| *DFT* | Deep Flow Technique |
| *pH* | Potential of hydrogen |
| *EC* | Electrical conductivity |
| *TDS* | Total Dissolved solvents |
| *NFT* | Nutrient Film technique |
| *ROI* | Return on investment |
| *CLV* | Customer lifetime value |
| *WOM* | Word of mouth marketing |
| *FPP* | Financial pain point |
| *Pr. PP* | Process pain point |
| *PPP* | Productivity pain point |
| *SPP* | Support pain point |

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# 1 Introduction

## 1.1 Background and problem description

The world's population is increasing and according to the United Nations, this number will continue to reach over 8.5 billion by the year 2050 [1]. One of the major challenges of this rapid growth is how to feed this ever-growing population. The amount of arable land – land that can be used for agricultural purposes – does not increase in step with the population. While the population is expected to grow by another 29% by 2050, the amount of arable land is expected to increase by less than 5% over the same period [2]. The main challenge here will be food production to this large quantity of the population and there's no escaping the fact that food production around the world needs to increase. This will shape the future of the farming industry in ways we haven't seen before, especially indoor farming where it provides the plants, additional nutritional value. The nutrition level increases as the plants are grown in a nutrient-rich environment and also due to the time saved in the transport of the plants, from the farm to the customer as they are grown in city spaces. This could be a positive development in the long run. With more than two-thirds of the world's population projected to live in cities by 2050. In-door farming will become more crucial as it requires no soil and less water than conventional farming. Indoor farming is a way for urban dwellers to grow their food, or at least have access to local food.

However, apart from the many benefits, there are problems with indoor farming as well. The research available refers to all the problems with indoor farming but does not address the people who grow these plants i.e., the indoor farmers. So, an exploration of the problems faced by indoor farmers is important. There can be solutions found to solve indoor farming problems in general, but how are these problems affecting indoor farmers is equally important.

## 1.2 Purpose

The purpose of the project was to understand what challenges and pain points, an indoor farmer faces during the whole growing process and how different criteria, activities, and different stages of the growing process play a part in their growing experience. There will also be an exploration of their thoughts/opinions on outsourcing technological expertise assisting the farming operations.

## 1.3 Context

This is a master thesis conducted within the master of Integrated product design at the KTH Royal Institute of Technology in Stockholm. This thesis was made in collaboration with an Agri-tech company. For client confidentiality reasons, the company will be called 'Green AB'. Green AB, an Agri-tech company, believes that indoor farming has lots of potential; and this technology, which has been growing during the Pandemic all around the world, can do wonders even during Post Covid circumstances to increase global food demands. Green AB is interested in this area and wants to investigate more on the user experience of the indoor farmers.

## 1.4 Research questions

1 What are the day-to-day activities of the indoor farmers in the growing process?
    a. How are they affecting the indoor farmers in the whole process?
2 What are the challenges and pain points faced by farmers?
    a. How are the challenges and pain points, different for small and large-scale farmers?
3 How can suggested solutions help the farmers' needs?

## 1.5 Deliverables

- Understand what are the challenges and pain points
- Quantitative analysis to identify the pain points of the in-door farmers.
- Qualitative analysis to understand the day-to-day activities of the farmers.
- Customer journey maps of the growing process.
- Concepts that incorporate solutions to the farmers' needs.

## 1.6 Limitations

- The time period provided for the thesis was 22 weeks.
- The thesis will focus was only on commercial indoor farmers who are growing leafy greens herbs and microgreens.
- The project was not location-specific to Sweden. Interviews of farmers and farm visits outside of Stockholm were conducted online.

- The research will focus on only the problems during growing.
- Due to time constraints the project proceeds until the User tests. The prototypes will be handed to Green AB for future testing and development.

# 2 Methodology

## 2.1 Overview

Human-Centered Design (HCD), is the method opted for this project. IDEO defines HCD as a creative approach to problem-solving that starts with people and ends with innovative solutions that are tailor-made to suit their needs [3]. This method will help the project understand the patterns, values, and needs of the indoor farmer.

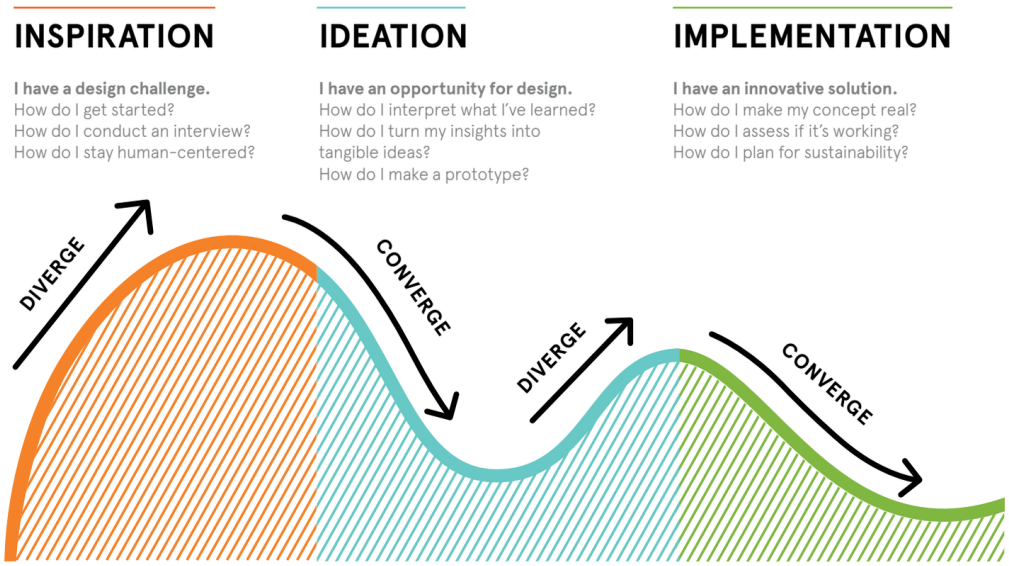



Figure 1. Human-Centered Design, popularized by IDEO.

## 2.2 Process

The Project process is visualized in figure 2. This process is tweaked and will overlay the design thinking process which will guide the research to focus more on understanding the user and creating impactful ideas. The beginning of the process appears to be a linear process, that is not the case and it is iterative in practice.

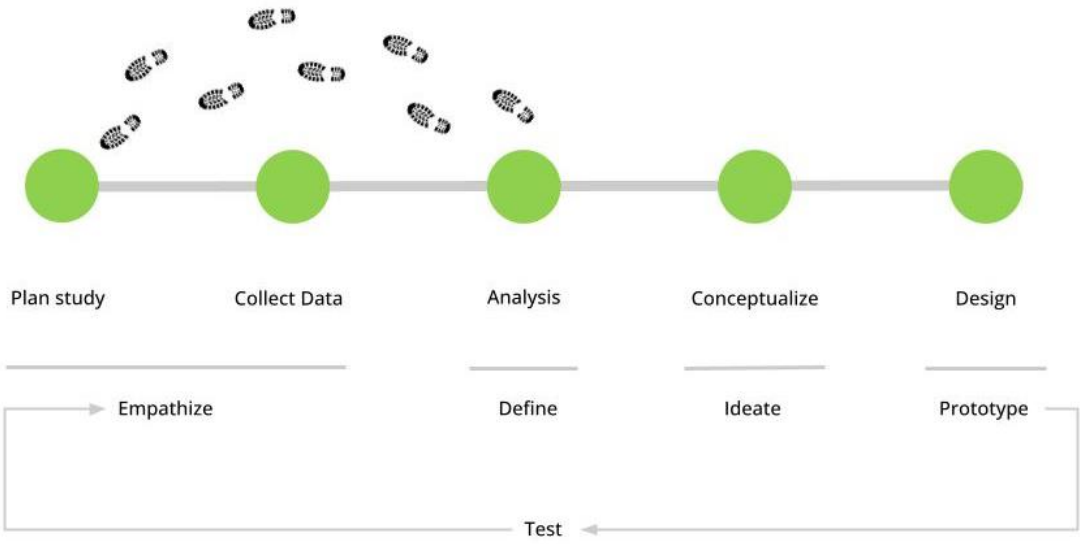



Figure 2. Project process.

## Background study

This project started with planning the research and what ought to be done in 20 weeks. A literature study was conducted as background research. This study consisted of a basic understanding of indoor farming and its methods; a brief of these methods is explained in the chapter *Background research*. Most articles found on Google scholar, Introduced indoor farming and its benefits. Sadly, a handful of articles provided information about the problems of indoor growing. To access more of this information, Blogposts, News articles, and YouTube videos were helpful. Since the research cannot be based on these subjective sources, a mind map was created after the background study to gain a broader overview of the problems of indoor farmers and they were noted down using keywords to be further validated.

The **user research** is mainly based on Qualitative and Quantitative research. The Qualitative research relied on methodological triangulation to gather data. Norman. K. Denzin presented this triangulation framework which involved using more than one research method such as a combination of In-person interviews, observations, and questionaries [4] to gain deeper knowledge.

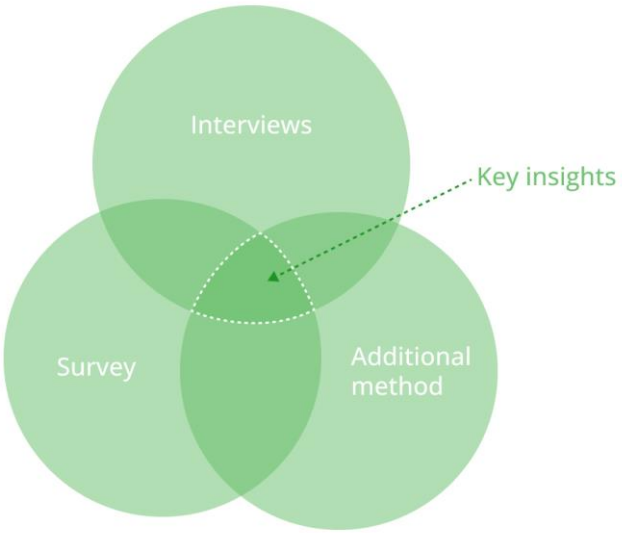



Figure 3. Methodological triangulation.

The mind map acted as a basic structure to create the questionnaire for the in-depth interviews that will provide Qualitative data. The interview format is a mix of questions and observations of the farm. Because the interviews were conducted at their farms, the interviewees can easily reflect on specific experiences and details, that would have been otherwise, missed.

User Journey map, leading to the pain points of an indoor farmer is explained in the chapter *User studies.* An additional method, try it yourself, was performed to understand the everyday problems.

For the Quantitative data, a survey was created based on the pain points, this is helpful as the survey will focus on quantifying specific pain points mapped out in the Journey map.

**Analysis**

The most important outcome of this thesis will be the insights, Farm visits, additional method, Qualitative & Quantitative analysis collectively form the key insights. A commonly used method called clustering is used in the thesis for analysis.

**Ideation & user tests**

'How might we questions' will be the basis of the concepts that were tested and further recommendations are presented in the chapter *Ideation.*

# 3 Background research

## 3.1 Indoor farming in general

Indoor farming also known as controlled-environment agriculture (CEA) is a method of growing plants, entirely indoors [5]. This method of farming often implements growing methods such as hydroponics, aquaponics and so, which utilize natural/artificial lights to provide plants with the nutrients and light levels required for plant growth. The main reason to grow indoors is to eliminate the use of soil. The soil in traditional farming provides two purposes, it acts as a support to hold the plants and it absorbs the nutrients and water needed for that plant to grow [6]. The issue is not all the water will be reaching the roots some amount gets evaporated resulting in more water usage. Indoor farming solves this issue by using less water and an additional benefit is that the plants are grown near to the consumer.

A wide variety of plants can be grown indoors, microgreens, leafy greens, herbs, fruit, and vegetables are the most popular. Indoor farming, often referred to as indoor gardening is being used to help strengthen local food supplies and provide fresh produce to communities in large cities [32] [5]. Many of these farms can produce much more plants on a small scale than produced through soil-based conventional farming. However, not all indoor farms are built on a small scale. Some large-scale farms can be created in an area as small as a basement and used by a single gardener to provide fresh produce throughout the year.

When growing indoors, many indoor farmers appreciate having more control over the environment than they do when using traditional farming methods. Light amounts, nutrition levels, and moisture levels can all be controlled by the farmer when they are growing plants solely indoors. Although growing plants indoors can limit growing options, gardeners and farmers have a wide variety of plants to choose from when deciding what to grow indoors.

Most indoor farming uses a combination of hydroponics and artificial lighting to provide plants with the nutrients and light they would only receive when grown outdoors. However, some indoor farming methods, like those implemented in greenhouses, can use a combination of natural and artificial resources.

## 3.2 Methods of indoor farming.

There are several techniques used to do indoor farming such as:

1. **Hydroponics**

Hydroponics is a type of horticulture that involves growing plants without soil, by using mineral nutrient solutions added to the water. Plants grow with their roots exposed to nutrient-rich water. The nutrients used in hydroponic systems can come from many different sources, including fish excrement, duck manure, purchased chemical fertilizers, or artificial nutrient solutions. Plants commonly grown hydroponically, include tomatoes, peppers, cucumbers, strawberries, leafy greens, microgreens, cannabis, etc. Lettuce is considered a beginner plant in hydroponics.

Hydroponics offers many advantages, notably a decrease in water usage in agriculture. To grow 1 kilogram of tomatoes using intensive farming methods requires 400 liters of water: using hydroponics, 70 liters; and only 20 liters using aeroponics [7]. Since hydroponics takes much less water to grow to produce, it could be possible in the future, for people in harsh environments with little accessible water to grow their food.

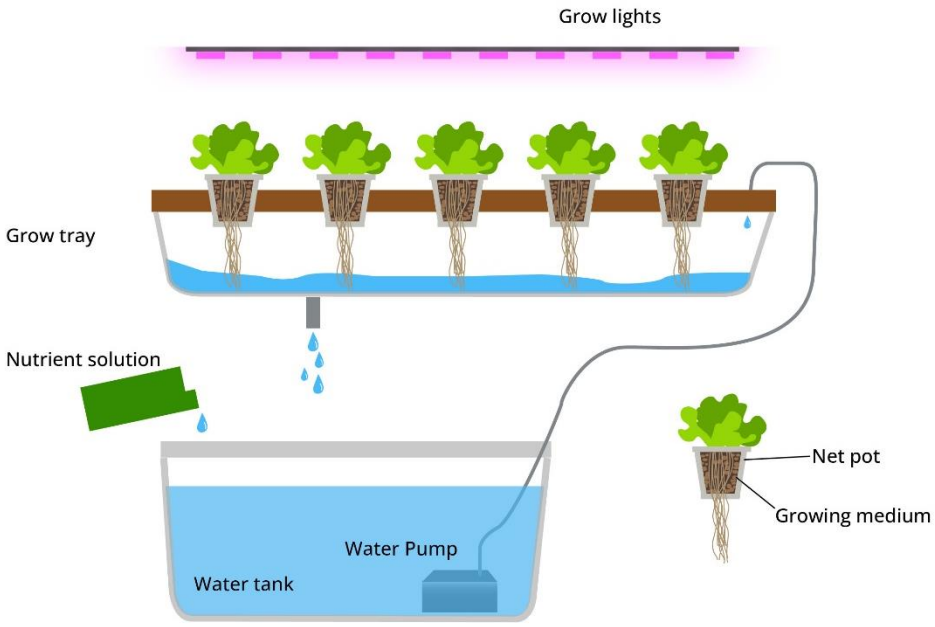



Figure 4. Hydroponic components

The basic components required for a hydroponic system are, a growing medium, nutrient solution, Water pump, grow tray, Net pots, water tank, and grow lights.

The growing medium also known as a substrate holds the seeds and supports the plant's weight, it is the part where the roots grow. The growing medium role is to act like a sponge that holds the water to maintain the moisture and provides maximum exposure to the nutrients the plant needs. The growing mediums used are perlite, Rockwool, coco coir, etc. [8]. When choosing the growing medium, the factor to consider is the type of plant to grow. However, the main factor will be the type of system used to grow.




Figure 5. Different growing mediums used in hydroponics.

The nutrient solution is similar to fertilizers. There is no set formula for these nutrient solutions since each plant requires a different amount of nutrients.

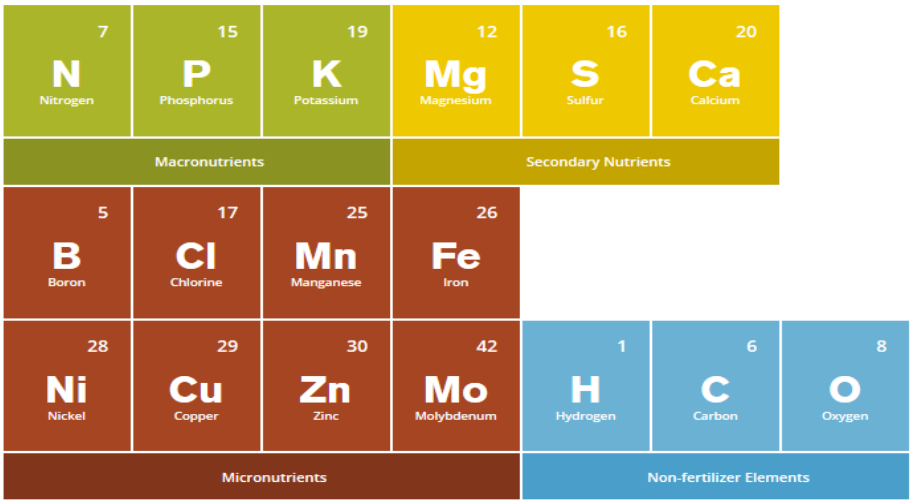



Figure 6. Periodic Table of Plant nutrients.

Basic nutrients for pant survival are oxygen, carbon, hydrogen, and nitrogen (O.C.H.N), all of these are readily available in the air. [9] Phosphorus helps in the development of flowering, fruits, seeds, and roots. Leaves turn yellow due to a lack of potassium. Too little calcium turns leaves brown. Magnesium helps create oxygen through photosynthesis and is often used in large amounts in fast-growing plants

The water pump is used to recirculate the water from the water tank to the grow tray. Basically, in DIY hydroponic setup nutrients are added manually in the water tank, however Commercial farmers with hefty investments, afford nutrient dozers, which provide precise amounts of nutrients in the water.

Net pot is a plastic structure that holds the growing medium that can be placed into the system.



Figure 7. Net pot (Left), a Net pot with rockwool as the growing medium (Right).

When it comes to Grow lights, fluorescent tube lights do work as grow lights but LEDs are widely used. In the light spectrum, red and white light are the two most important for photosynthesis [31] which makes LED light the ideal choice to go for.

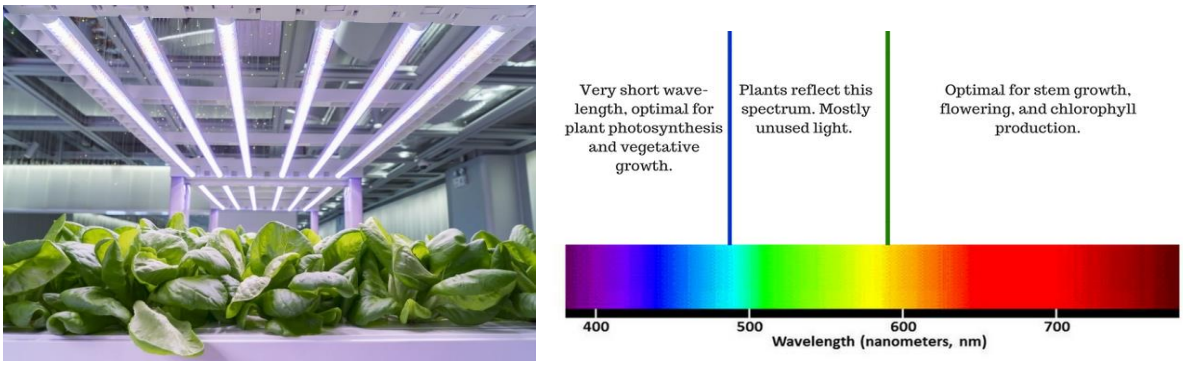



Figure 8. Placement of the grow lights to provide photosynthesis (Left), different Light spectrums for plant growth (Right).

## 2. Aeroponics

Aeroponics is a hydroponic method of growing plants in a mist environment without using soil. The basic principle of this technique is to spray nutrient-rich water solution to the roots of the plants at a consistent rate and time period. Basically, it is an air-water culture cultivation system, the roots of the plants are hung inside a sealed container under darkness and openly exposed in the air to get nutrient-rich water sprayed through atomizers. The upper portion of the plant leaves and crown, extend above the wet zone.

Ideally, the environment is kept pest-free so the plants grow healthier and quick. The benefits of this technique are that it delivers more oxygen to the plant roots with a plentiful supply of water and nutrients, stimulating growth [7] [10]. Figure 9 illustrates the side view of the system.

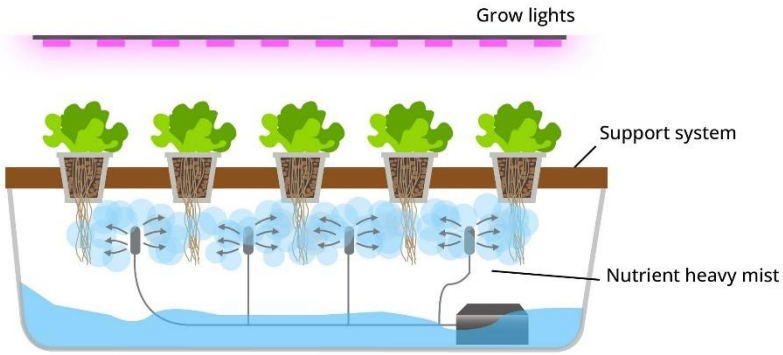



Figure 9. Side view of an aeroponic system.

The grow lights enable the plants to photosynthesize and store energy. A support system acts as a barrier that separates the plants from their roots to give them support to grow in the entire plant cycle. Finally, the solution chamber is where the nutrient-heavy mist is sprayed onto the roots.

### 3. Aquaponics

This is a growing method that uses aquaculture and hydroponics to grow plants. The aquaculture can be saltwater or freshwater fishes, prawns, and crayfish. The water used in aquaponics is recirculated and reused very effectively. Figure 10 explains the closed water system of the aquaponic system.

The water collects the ammonia generated by the excretion and gills of fishes and generates the nutrients to be fed to the plants. This condition is great for home aquaponics, but for commercial purposes and large-scale operations, there would be a nutrient deficit in iron and potassium which have to be added separately [10].

A few other micronutrients are added for better plant growth. The process is similar to the hydroponic system, instead of adding nutrients to the water the aquaculture excretion acts as one of the main nutrient resources. The main disadvantages are, the initial setup cost is high and it requires high maintenance of the aquaculture.

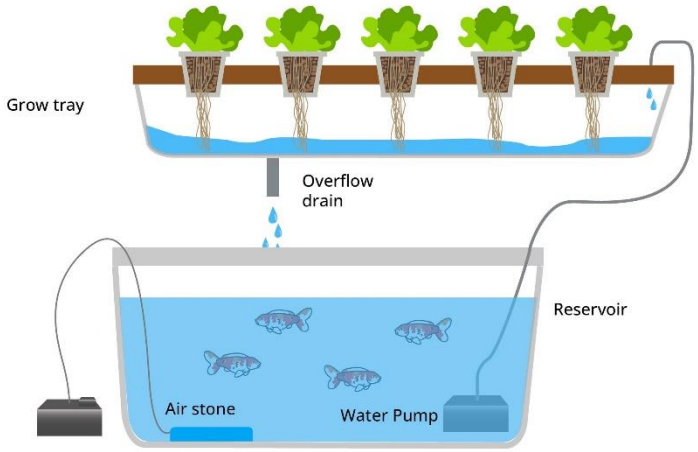



Figure 10. Aquaponic setup.

The different techniques of hydroponics are as such:

**Deepwater Culture (DWC)**

DWC, also known as Deep Flow Technique (DFT), is a hydroponic growing method where the plant roots are suspended in a solution of nutrient-rich, oxygenated water. This method uses a water tank less than 0.5 m deep filled with a nutrient-rich solution with plants floating in a support system on top. Typically, DWC is used to grow short-term plants, ideally, leafy greens and herbs. The water helps decrease rapid changes in temperature, pH, electrical conductivity (EC), and nutrient solution composition [11]. The disadvantage of this method is that diseases can spread quickly which means the transfer of pathogens from one plant to another. Figure 11 demonstrates the placement of the plants in an oxygenated, nutrient-rich water tank.

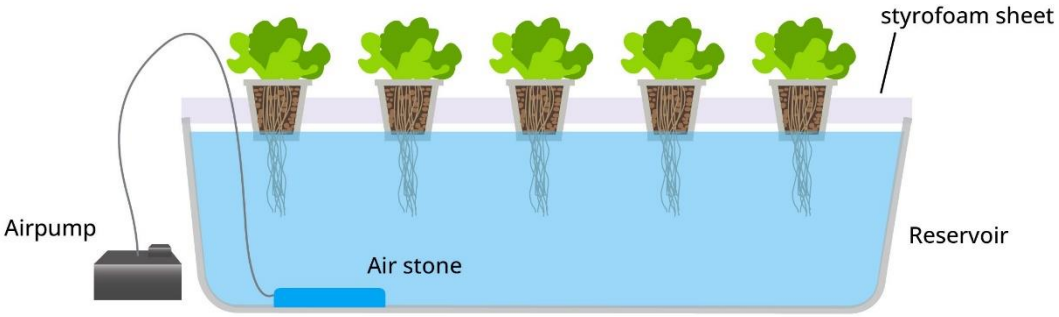



Figure 11. Basic DWC setup.

**Ebb & Flow (flood and drain) sub-irrigation**

Ebb and flow consist of two stages, Ebb is the outgoing stage when the water is drained out from the tray and Flow is when the water is filled back into the tray. The benefit of the system is that the solution is not left in constant contact with

the roots it provides more passive oxygenation which tends to suppress pathogen growth [12]. Figure 12 explains how the system is set up.

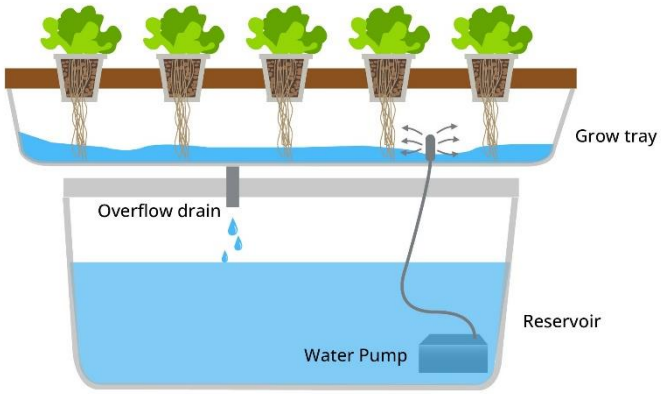



Figure 12. Ebb & Flow system.

In its simplest form, there is a tray above a reservoir of nutrient solution. Either the tray is filled with growing medium, clay granules being the most common, or in an individual net pot containing growing medium. At regular intervals, a simple timer causes a pump to fill the grow tray with nutrient solution, after which the solution drains back down into the reservoir. This keeps the growing medium regularly flushed with nutrients and air. Once the upper tray fills past the drain stop, it begins recirculating the water until the timer turns the pump off, and the water in the upper tray drains back into the reservoirs. There are certain drawbacks to this system; it is labor and energy-intensive in large-scale applications and sensible for growing large plants. The second drawback is the pH of the nutrient, fluctuates and if pH is not corrected various problems may occur affecting the plant.

**Nutrient film technique (NFT).**

The nutrient film technique is a hydroponic technique wherein a shallow stream of water is recirculated in channels to the plant roots. The design of this technique provides an adequate supply of nutrient-rich water and, oxygen to the roots. Figure 13 explains a typical NFT setting that is used both in a large and small-scale indoor farm.

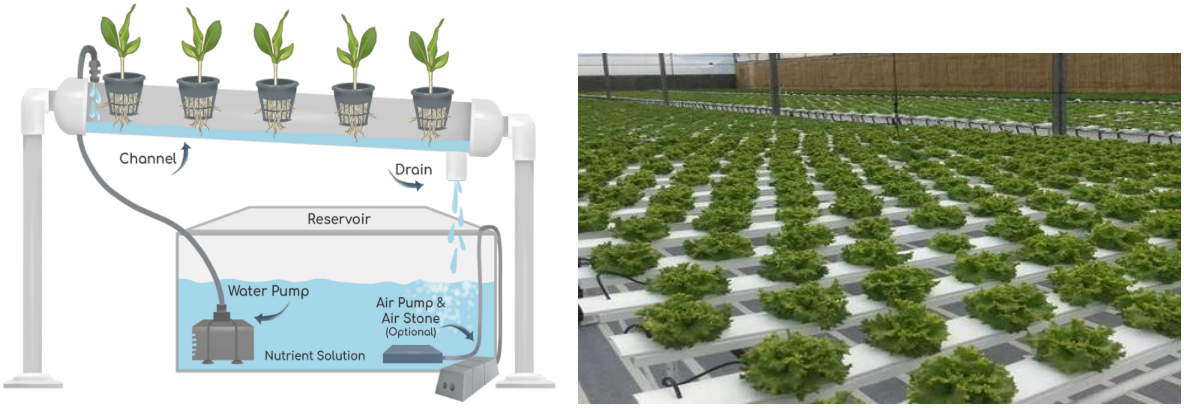



Figure 13. A typical NFT setup (Left), Greenhouse Farm with NFT setup (Right).

This setup is multiplied and the number of Channels is lined to achieve maximum yield. The basic principle is that a constant rate of nutrient-water mixture is passed through the channel needed for the roots which creates more oxygen present in those channels, this allows three requirements to be met at the same time for healthy plant growth [31][32].

### Drip irrigation system

A drip irrigation system is an active hydroponic system. This means that it uses a pump to feed the plants with nutrients and water regularly. Such a system is widely used in outdoor gardens to deliver water and nutrients to individual plants. The system uses a network of feeder lines to deliver the water to the plants. This is best suited for large growing operations.

The system uses individual pots for plants. The water from the reservoir is connected to the plants by a network of tubing. Individual crop gets at least one dedicated drip emitter. Each emitter has a mechanism that allows controlling the flow of water. The flow to the plants has to be regulated in a drip system [31]. The growing media needs to be given time to breathe in between flows. If left uncontrolled, a drip system will flood the plants and eventually drown them. So, all drip systems use some kind of timer system to regulate the flow of water and nutrients to the plants.

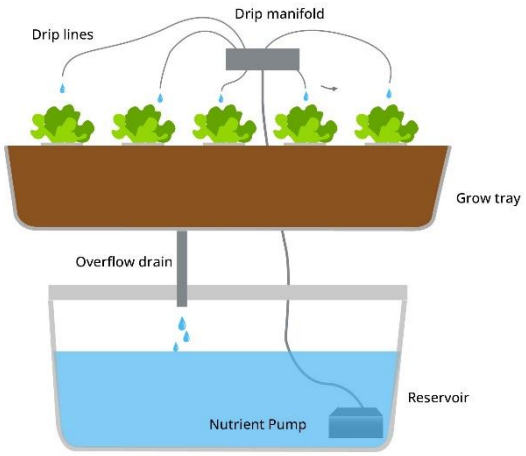



Figure 14. Drip irrigation setup.

In typical situations, the pump is operated several times a day to send water to the plants. Such a system requires considerable planning and effort in the initial stage. But once the drip lines and carefully installed, the system can run with minimal assistance. The disadvantage lies in the complexity of small grow operations. The advantages of this system are that it provides more control over water, nutrient supply, and low maintenance compared to other methods in addition to the variety of deep root plants that can be grown. Table 1 shows the techniques that will suit different plants [28]. The types of crops can be divided into three phases:

Phase 1- Leafy greens, microgreens & herbs.

Phase 2- Vegetables, fruits & roots.

Phase 3- Staple crops.

Table 1. Shows what plants can be grown successfully in a particular growing method [28].

| Type of system | Type of crop |
|---|---|
| NFT system | Phase 1 crops (Leafy greens, microgreens & herbs) |
| EBB & flow system | Phase 1 & 2 crops (Leafy greens, microgreens, herbs, vegetables, fruits & roots) |
| Deepwater cultivation | Phase 1 crops (Leafy greens, microgreens & herbs) |
| Drip irrigation system | Phase 2 crops (vegetables, fruits & roots) |

All though this might not be the ideal case. Improvisations and new growing materials can help grow any phase of crops into any type of hydroponic technique.

### Tower farming - ZipGrow™ towers

Tower farming is a type of technique suitable that uses modular systems to grow in urban environments. ZipGrow™ Towers are such modular systems, these growing towers can be moved and taken apart easily, allowing users to grow extremely high-density vertical farms that are also convenient for labor.




Figure 15. Greenhouse using ZipGrow™ tower (Left), ZipGrow™ tower setup (Right).

The container consists of one strip of matrix media slid into the housing. The matrix media acts as a support that holds the plants. Before placing the media into the housing, the seedlings are transplanted into them [14] [15]. Once the seedlings are placed, the housing is hung vertically and water drips down due to gravity. Figure 14 shows these housings being placed vertically. The advantages are that it's a one-time investment as these containers and media matrix can last for a long time. Another advantage is the reachability to every plant is equal compared to horizontally grown systems.

### Vertical farming

Vertical farming is a practice of Growing plants and stacking them vertically. This is a great example of increasing crop yield with a small land area. This can produce 10xyield compared to traditional farming. Abandoned buildings are often reused for vertical farming.



Figure 16. Vertical farm. Source: Medium | AeroFarms.

Some claimed, vertical farming technology is the future of agriculture, promising a more sustainable approach to growing, high quality, healthy produce, and while many challenges remained, it certainly seemed like this technology was already an effective way to grow leafy greens and herbs, and even showed promise towards growing more challenging, but impactful crop types in the future [31] [26].

**Plant cycle.**

The plant cycle consists of different growth stages from seed, baby leaves to a mature plant. This includes the number of days the plant needs to germinate and develop leaves and reach maturity so it can be harvested. Plant cycles have to do with a lot of factors including the specific type of plant or variety being grown [27] [32]. There isn't any specific number of days that can be expected, as there are more complexities of the process. For instance, hydroponics growers have been able to induce early flowering in plants by changing the light cycles in a grow area.

One of the best things about growing indoors is what it has to offer in terms of time management. With traditional soil-based growing, farmers grow from seeds, they have to have that plant in the same patch of soil for its entire life cycle [16]. That means they have to dedicate the whole maturity cycle for that area. So, for example, a basil plant takes 28 days to mature. The grower is going to have a harvest 28 days after they put that seed in place. Taking that same 28-day plan to a hydroponic setup. They will start germinating the seed in an indoor hydroponic setup, which might take around 7 to 10 days. Then transplant the seed to their main growing area. So, in the growing area, they have another 21 days to harvest. The reason for doing this is to refill the germination stage trays while the other plants are growing and cut 7 days off the cycle.

Multiplying this process by another couple of stages, we can see how growers can generally harvest every 10 or 14 days. It's not about speeding up plant development, it's about handling multiple stages of plants in a grow area. This handling of multiple stages of the plant can be challenging and there is an important task involved of maintaining the system properly and keeping the plants safe. There is a huge risk of losing a plant and mainly to diseases, which are responsible for what can go wrong with the plants. A few common indoor plant diseases are as such:

- Powdery Mildew – It looks like sprinkled white powder over the leaves and stems and if left untreated it will stunt plant growth, leaf drop, and yellowing of plant tissue. If it gets too far, the plant has to be discarded.
- Downy Mildew – It mostly appears on the underside of leaves and doesn't look like a powder the way "powdery" mildew does. They both can cause yellowing of leaves which makes them easy to mis-identify though.
- Gray Mold – Also called ash mold & ghost spot; it starts as spots on leaves that lead to fuzzy gray abrasions and will continue deteriorating until the plants are brown and mushy.
- Root Rot – When there is too much water and pathogens in the medium, there is a high chance to get root rots. Plants will wilt and turn yellow. Roots can get mushy too.
- Iron Deficiency – Plants lacking iron will lack chlorophyll, the leaves turn bright yellow while retaining green veins. Sometimes this is misdiagnosed as a different type of disease when the plants are simply lacking iron.

Though these diseases are a problem to every indoor farmer, another such threat is a pest, which can cause damage and undo all the work in the Farm/Greenhouse. A few common indoor plant pests are as such:

- Spider Mites – these tiny spiders (less than 1 mm long) are probably the most common of all indoor garden pests. They are little arachnids and because of their small size, it is hard to notice them until they do serious damage to the plants.
- Thrips – are tiny (around 5 mm long). Though tough to identify, their damage is easy to see. These are small metallic black specks on the top of the leaves. This will often be accompanied by the leaves turning brown and dry (possibly with yellowish spots) because the thrips have sucked the leaves dry.
- Aphids – also known as plant lice. They can be green, black, or gray. No matter what color, they can weaken the plants because they suck the juice

out of leaves and turn them yellow. They may be found anywhere on the plant, but they often gather around the stems.

- Whiteflies – are about 1 mm long and look like small, white moths. This makes them easy to spot, but they're harder to kill because they fly away when disturbed, to a new plant. They also suck the plants dry causing white spots and yellowing.
- Fungus Gnats – surprisingly, the adult fungus gnats are not harmful, but their larvae feed on roots and feeder roots, which can slow plant growth, invite bacterial infection, and if taken to the extreme, even plant death.




Figure 17. Tip burnings, root rots, leaf withering, and pests |Facebook groups and Reddit posts.

## 3.3 The Business of indoor farming.

Although the main purpose of the thesis is to identify challenges and pain points during the growing process, it is equally important to look into and have an overview of the business as well. To understand the business better, a hypothetical business flow was created based on the references [29] [31], Figure 18. This flow depicts how the business works and how different tasks take place.

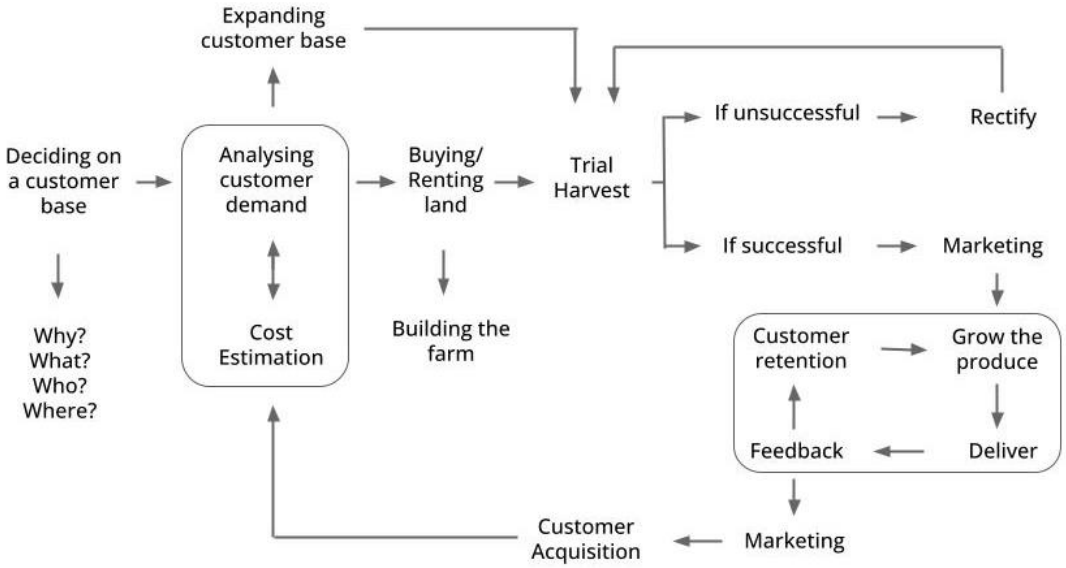



Figure 18. The business flow of indoor farming.

This business flow can vary depending upon the availability of land to build the farm or the goal to supply the customer base. The customer base is the food communities that encourage the idea of indoor farms and are potential repeatable customers. These customers can be the main source of income to the farmers and are the initial target audience.

During the background research, there was speculation of three aspects from the business flow that were considered important to an indoor farmer.

**Customer demands**

The laws of Supply and demand very much affect businesses in general. To ensure that a great idea has the potential to sustain itself, the customer demands need to be evaluated regularly.

The industries respond to market demands which is a collective of individual customer demand. This demand is basically for fresh produce and locally grown all over the world [29] [17]. Market demand in indoor farming describes the demand for a given crop and who wants to purchase it. This is determined by how

willing consumers are to spend a certain price. When the demand decreases, the price will go down as well. Market demand is the total of what everyone within a specific industry desire and can help guide Indoor farmers when growing the plants. See Appendix B for customer demand analysis of a mid-scale indoor farm from India.

The farmers have invested huge amounts into the business and for them, return on investment (ROI) is important. The real competition is traditional farming as they are linked to our supply chain which has been improving over decades [26]. The main task for farmers to gain profits would be to grow in terms of customers, by retaining previous customers and bringing in new customers.

**Customer Acquisition and Retention**

The two key drivers that help drive growth in any business are customer acquisition and customer retention. Customer Acquisition involves processes put in place for the farm to bring new customers on board and convince them to purchase the product. The customer base expands as a result of the acquisition. On the other hand, customer retention is about keeping the initial customers and holding on to them to increase sales and profit, this is to encourage them to repeat their purchases by maintaining brand loyalty.

The cost of acquisition is higher for an indoor farm and it may go up to five times more than retention, as with a new customer it is similar to starting from scratch and needing to plant the seed all over again. A good customer retention strategy allows the indoor farmer to improve their customer lifetime value (CLV) by encouraging them to return over and over again by providing high-quality produce. Retention has a higher Return on investment (ROI) and is the key to making indoor farms profitable. The advantage of retention is Word of mouth marketing (WOM) is one of the most effective forms of advertising [18]. Customers who have brought indoor-grown produce for a long time are more inclined to suggest the product to their family and friends. All in all, Customer retention is important for long-term sustainability while acquisition is critical for long-term expansion and helps balance the customers and retain the revenue.

## 3.4 Mind map

A detailed view of problems and challenges were broken down into a mind map. This was made to gain a broader overview of the inside information of an indoor farm. The mind map summarized the reasons for growing problems, and various connections from growing methods to identifying diseases were made, that were needed from the background study. Table 2 shows the different categories generated during the process. See Appendix A for a detailed view of the mind map.

Table 2. shows the categories that were identified.

| | |
|---|---|
| Location of the farm | Plant monitoring |
| Growing systems | Initial costs |
| Diseases | Plant chosen |
| Knowledge | Labor |
| Customers | Component failure |
| Maintenance | Energy |
| Time-consuming | Business plan |

# 4 User studies

This research focuses on the challenges and pain points of the target audience. In this study, the target audience is Commercial indoor farmers, and to tone down the heterogeneity the study will focus on Commercial indoor farmers who grow Hydroponically, Vertically, and leafy greens/herbs, in a Controlled-environment or a Greenhouse.

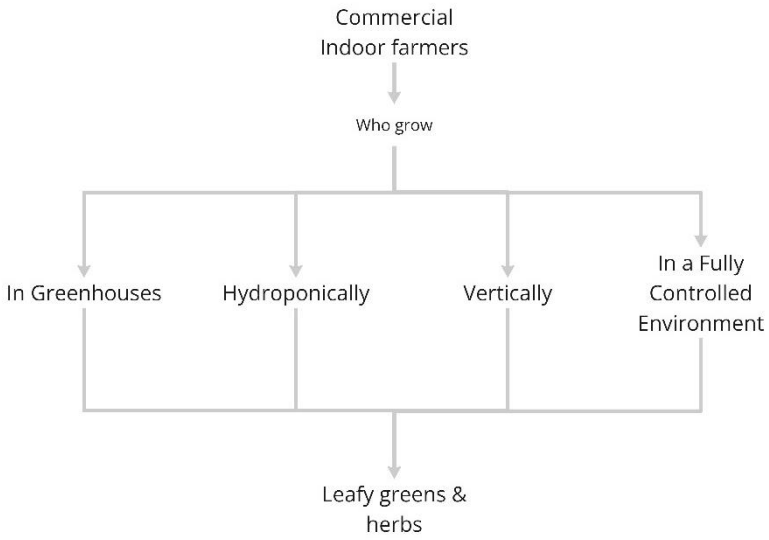



Figure 19. Target audience

## 4.1 User narration

Three indoor farms were visited irrespective of the geographic location as it was not the main focus of this study. The farm visits were to understand the user's environment and behavior towards their daily activities.

The first farm was in Stockholm in which the Lead grower was interviewed. The farm is a ~600 sq meter hydroponic farm that uses **ZipGrow™** towers, to grow the produce vertically because of which the farm can harvest what a 3000 sq meter farm can harvest. The interviewee works as a plant scientist along with 6 employees who manage the farm to help with the day-to-day tasks. The research team updates the interviewee on the market demands and what plants need to be grown. The week starts by seeding which is the first task of the plant cycle. The interviewee expects the employees to be active as the farm harvests leafy greens every week. These harvests are planned in such a way that after a tower is harvested a new batch of plants are transplanted simultaneously to start a new plant cycle. The interviewee usually spends around 8 hours on the farm. These 8 hours are spent on Quality Checks, R&D, and Pruning. The seeding, transplanting, and harvesting are done by the Interns (hired every quarter) and by the Research

assistants. Every day the interviewee assigns tasks to the employees based on the demand of that particular task and according to the competencies required. Since the interviewee qualifies as a Plant Scientist, they express huge responsibility towards Quality checks before the plants are harvested. The appearance of the plant matters, as the consumer, considers visually appealing leaves to be healthy. This has cultivated the idea of supplying premium plants to the customer which is of greater importance. Pest scouting on the plant leaves is done every alternative day and during scouting, the interviewee feels the task to be stressful and exhausting to look for tiny pests, figure 20. The farm wants to sell fresh produce only to supermarkets and restaurants in the Proximity of a 5km radius, which shows there is no interest in selling to customers outside of the Food and beverage industry.



Figure 20. Interviewees, inspecting the leaves in the First farm (Left) and demonstrating pruning and regular checks (Right).

The second farm visited was in Brno, the Czech Republic, the company-Future farming is using an 800 sq meter **Vertical farm** that uses aquaponics to grow fresh leafy greens and herbs to supply to the local community. This farm visit was conducted on a video call. The interviewee works as a grower along with 10 other employees who work 8-10 hours a day. The interviewee starts the week by seeding based on the customer demands. As soon as the plants reach a certain height, they are transplanted into the growing area. A regular day consists of tasks such as managing the nutrient levels, maintaining the humidity of the farm, manually reading them, disease checks of different batches of plants to ensure no abnormalities or spots appear on the leaves. Similar behavior to the first farm is observed here with two growers, to pick over the best leaves while harvesting, and this happens involuntarily. As mentioned earlier, the farm uses aquaponics to grow which means the biggest resource is the fish and this comes with its challenges. Aquaponics is labor-intensive as there needs to be consistency in maintaining the salinity of the fish tanks which is why apart from the growers there is a “fish team” to look after the health of the fishes. The interviewee expresses how the plants can manage without water supply for a day or two due

to the moisture present in the growing medium, but a large number of fish that are grown in large fish tanks, figure 21, makes them vulnerable to any type of water impurifications.

*"There needs to be a balance between the fishes and the pants."*




Figure 21. Fish tanks (Left), demonstration of the light intensities of the Container farm (Right).

The third farm visit was in Boston, and this was a **Container farm** [19]. Shipping containers are refurbished by several companies into self-contained vertical farms, complete with LED lights, compact growing systems that can be vertically or horizontally stacked to grow different plants. This farm uses a combination of drip irrigation and a Vertical tower system.

The farm is located under a freeway, and the interviewee who is a grower, explains their 5-hour work experience every alternate day. The farm is managed by two people with no educational backgrounds in horticulture. There has been a lot of empirical knowledge that is gained and this knowledge is the only reference in making growing decisions. Since this is a container and plants are grown in a controlled environment the interviewee explains biosecurity to be the utmost priority when entering the farm. Emphasis was on protective gear as a preventive measure to stop contamination through humans. Previous harvests had resulted to be unsuccessful because of contaminations. Due to ongoing demands for Pak choi, the interviewee was keen on growing them but faced a certain problem regarding light cycles. The light intensity of 5:2 red to blue ratio was maintained, 12 hours of light and 12 hours of darkness resulting in uneven growth of Choi leaves and was corrected by externally consulting for the next few harvests. The farm usually sells directly to customers on a subscription basis and the everyday concern for the interviewee is not being consistent in terms of Quality. The interviewee talks about saving expenses by consulting were beneficial until an IoT app was used for the farm The app alerts the interviewee when there is less nutrient concentration supplied to the plants.

For this project, the target audience is categorized based on their farm-scale: large scale (1000+ sq meters), mid-scale (500-900 sq meters), and small-scale farmers (less than 500 sq meters).

## 4.2 In-depth interviews

One of the main aims of the in-depth interviews was to be as diverse as possible in terms of the location to keep a broader perspective. Ten in-depth interviews were conducted. Every interview commonly lasted around 50-65 min. An interview guide was created to help structure the interview. See Appendix C for the interview guide. Two interviewees with Mid and Large-scale farms were briefed with a few questions about the Knowledge gap in Growing decisions and Wastage occurring during the plant cycle, prior to the interview for them to prepare and provide more accurate insights.

The interviews covered aspects such as everyday work, emotions pertaining to certain tasks, growing problems faced, lifestyle behaviors. The interviews helped the study to differentiate the types of indoor farmers, Pure-growers, and Providers. Pure-growers are farmers, whose sole purpose is to provide nutritious produce, diversify the plants, work on improving the Quality to supply the customer demand. Whereas Providers concentrate only on research by gathering plant data and different plant recipes to either sell them to or develop a product/technology, based on the plant data gathered for home growers and upcoming indoor farms. Though they grow plants and sell them to their customers, their business is focused on developing technology, based on the data gathered. Three interviewees were Providers, developing R&D models, home grower kits, hydroponic greenhouses, and the rest seven were Pure-growers attending to the regional customer demands. for customer demand analysis, see Appendix B.

The interviews provided the research with specific data about the challenges based on the farm scale and the day-to-day activities which helped to form a *journey map*.

After these ten interviews, the author reached saturation and decided to continue further with the analysis.

## 4.3 Qualitative analysis

After gathering the data from the interviews, it needed to be analyzed and a commonly used method called Clustering was used to find patterns and themes [20]. This method was started by writing down all the useful information, quotes on post-it notes. These post-its were grouped according to the interviews. Figure 22. These notes were color-coded to trace back to the interview from where this data was collected.




Figure 22. Color-coded post-its before clustering.

The post-its were clustered into themes and during this, certain post-its were filtered as they did not serve any purpose to the aim of the study. For close up images of the clustering see Appendix D. The themes formed were as such:

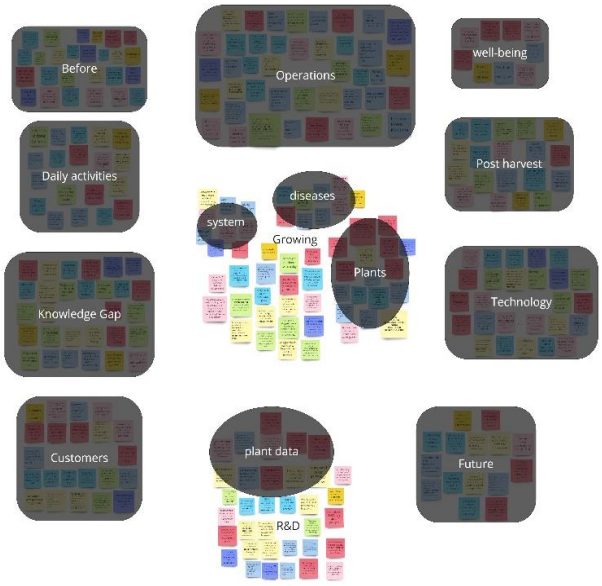



Figure 23. Clusters forming themes.

**Before**- Farmers described issues during setting up the farm and found it difficult to source growing materials.

- *I had a terrible time with my contractor.*
- *I changed vendors as the materials were not reliable.*

**Daily Activities** - Indoor Farmers generally put a lot of emphasis on the daily activities which include the monitoring of the plant and basic checkups- humidity, water & air temperature, EC, ph, nutrients, light intensities, and disease checks.

- *Regular upkeep of the farm. You just be there in case something doesn't work.*

**Operations** – This theme consists of the internal processes of the Farm/Greenhouse. Tasks such as seeding and transplanting need manual labor. Farmers talk about their experiences in managing diseases and plant health.

- *The main manual work is mostly seeding and transplanting, and harvesting.*
- *I introduce ladybugs to combat bad insects!*
- *Hired a person with Experience in Greenhouse because of the similarity to help with the growing*

**Well-being** - The farm environment creates a very soothing and calm environment to work in and the farmers feel relaxed after work. Disease checks are exhausting and the task causes stress while looking for minute pests.

- *It's a very positive environment to be in.*
- *Visual inspection of leaves is exhausting*

**Customers** – The farmers describe who their customers are and the needs of their customers.

- *My customers want traceability.*
- *We are pure-growers and also providers of the NFT system as a service.*
- *People found the papayas better than store brought papayas, So I'm growing them.*

**Post-harvest** – The theme consists of insights regarding problems after harvesting especially wastage observed by the farmers.

- *Most of the wastage happens during harvesting because the leaves are mishandled and crushed.*

**Growing** – Most of the farmers had negative experiences with growing because of diseases, the components used in the system, and the plants that were chosen to grow. Since they had very little experience with new plants most fail to grow plants successfully.

- *Unsuccessful germinations.*
- *We failed to grow strawberries, potatoes, iceberg lettuce, and more.*
- *We failed to maintain the salinity of the fish water and high pH is not good for the plants.*

**R&D** – The theme concentrates on the insights which highlight the farmers' reason for doing R&D. This includes the pure-growers and the providers.

- *What is widely consumed in India & how can they be grown.*
- *I did not spend time growing tomatoes because it didn't provide any scientific data!*

**Technology** – Farmers are interested in adopting technology for their farm but strongly express that buying tech is not an option considering the initial investment of the farm and also a trusted company is important in buying technology. Farmers express technologies like harvesting & transplanting machines, and IoT systems to be helpful in growing activities.

- *The initial investment of the farm makes the availability/opting of new tech less likely.*
- *" I'm willing to outsource tech which is in 'plug n play' condition"*

**Knowledge Gap** – Since indoor farming is still in the newest form farmers have not gained enough experience in growing different plants. The knowledge to know what to use to grow plants is still unmarked boundaries and also there is still the question will people choose indoor-grown produce.

- *Knowledge about Hydroponics/indoor farming is fairly low.*
- *India is perpetually following conventional farming.*

**Future** – Farmers describe what the business wants in the upcoming years as well as the resources they would need to solve their current challenges and problems.

- *Investment in tech leads to less human interference.*
- *Workers are the most important resource but we also want to invest in terms of space, time, money & education*

### Farmer's context

Lettuce is known as a beginner plant in indoor farming and this is easy to achieve with a lot fewer resources to be used. Growing only lettuce doesn't account for a healthy ROI, so they start with the next easy plant to grow which is in high demand. Apart from lettuce, this trial plant is where the growing problems also occur. Nutrients are directly associated with the flavor of the plant [interviewee 3&5]. Due to the different nutrient requirements of the new plants, they start to observe tip burnings and withering of leaves. Not having enough research data on the plant to grow them successfully is observed in several interviews and it worries most of the indoor farmers. They have a need to grow different plants to retain their original customers. Farmers who manage to grow different plants such as. Lettuce, basil, kale, Pak choi and so, successfully, feel relieved and are most likely to concentrate on acquiring new customers and attending to their demands.

It is found that indoor farmers believe, customers are more likely to buy plants/products that are visually appealing, "premium, healthy-looking" plants. This has influenced their decisions and are harvesting based on the appearance and volume of the plant. Blemishes on the leaves are unacceptable. So, this leads to crucial user tasks such as consulting and inspecting the plants regularly.

The amount of information that is required to solve the growing problems exceeds certain small-scale and upcoming farmers' abilities; the overall farm efficiencies suffer. The reason can be speculated to their attitude "anyone can do this type of farming" and, educational qualifications, but the number of interviews cannot surely determine this.

A standard practice of checking the plants and nutrient levels is done on a daily basis. Along with that, extensive cleaning of the grow areas after every harvest to create a healthier environment for the plants and, making it as sterile as possible.

### Needs and values

After the clustering, each interview was analyzed to understand what are their needs and what do they value. To determine the needs and values, The Element of Value pyramid by Harvard Business Review was used as a reference [24]. An article from UX collective states that the value definition is entirely intrinsic to users and when the product or service delivers these user values, it provides a better experience [25].

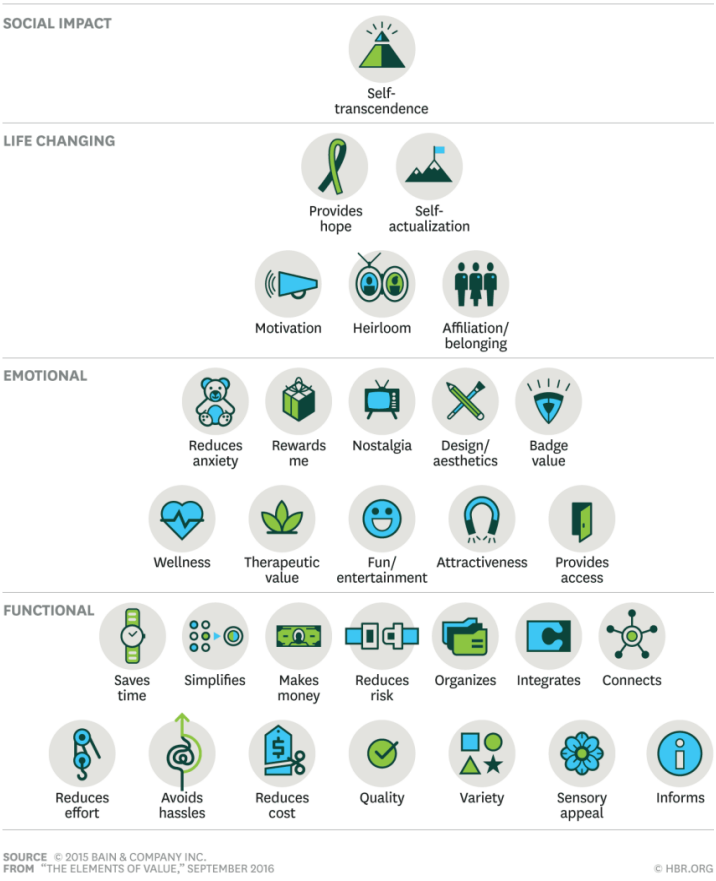



Figure 24. The elements of value [24].

The element of value is an extension of Maslow's hierarchy of needs, figure 24 refers to the elements, the solutions need to deliver in order to improve the experience.

Analyzing every interview helped to derive certain user values explained in table 3.

Table 3. User values

| | |
|---|---|
| Functional needs | Make money, Reduce effort, Save time, Sense of comfort, Simplifies. |
| Emotional needs | Provide access to information. |
| Life-changing needs | Create motivation. |

## 4.4 Additional method

### Try it yourself

This method allows the researcher to experience and develop the awareness of the end-user by using the product or service [19]. This method was performed parallelly when conducting the in-depth interviews to empathize with the indoor farmers and create a deep understanding of the growing problems. A small farm space was utilized, which already had a functional Deep-water cultivation (DWC) system. Lettuce was the choice of plant to grow considering no previous knowledge in this field. Figure 25, shows all the tasks performed.



Figure 25. Demonstrating the tasks performed- Seeding, Seedling humidification, Transplanting, Harvesting, and Packaging.

After a week of seeding, seedlings appear, and achieving 100% germination was unsuccessful. Figure 26 shows not all the seeds have sprouted. The issue was observed in most of the in-depth interviews. An evoking thought is that the DWC setup has the capacity to grow 30 plants.



Figure 26. Loss of germination.

After transplanting the water was manually checked every alternate day, to ensure the nutrient levels and pH levels are stable with the help of a pH meter and electrical conductivity meter [30].

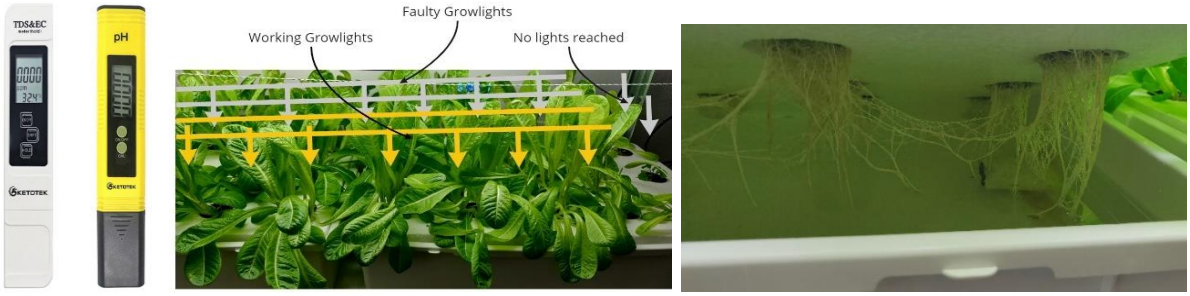



Figure 27. pH and TDS/Electrical conductivity meter (from the left), lighting problem experienced Roots inspection.

After 2.5 weeks the plants started to reach a certain height and produced volume. During the 3rd week, two out of the four grow lights (CFL tube lights) stopped working and adding to that, the plants were not placed exactly below the grow lights, due to which the plants grew taller with thinner leaves and long stems. Figure 27 demonstrates the situation. A similar situation was observed in two in-depth interviews.

The roots were checked regularly during maintenance, for any signs of root rots. To perform this, the whole support platform was lifted for better visibility, in addition, there was difficulty to properly examine the roots at the far end.

A few concerning issues were, yellow spots appearing on the leaves, molds forming on the growing medium, and withering leaves. Figure 28 shows these deformations, along with the wastage due to pruning these deformations.



Figure 28. Demonstrates growing problems-blemishes on leaves, withering, lighting problems, molds & more wastage.

Try it yourself was an insightful method and during this method, tasks such as seeding, checking the plants regularly, harvesting is where the author experienced stress and anxiety-free states and a sense of freedom to escape the background noise of thoughts and feelings.

All in all, in comparison to the DWC setup size, for an average scale indoor farmer with customer demand for diverse plants, these issues would be concerning as the efficiency of the farm would decrease. These are noted as significant growing problems and considered informing the key insights.

## 4.5 Journey mapping

The main outcome of the Qualitative analysis was to identify what are the day-to-day activities. This was useful to map out the User Journey [22]. The journey map was mainly used to find out the pain points and opportunity spaces in the process. The map shows the journey and emotions connected to the stages and also includes insights collected from the interviews. Pain points are problems and they're often grouped into several categories [23]. Categorizing the pain points helps to highlight the opportunity space associated with the context of the problem.

The pain points are categorized into four types:

- Financial Pain Point (FPP) is where the user wants to reduce their spending.
- Process Pain Point (Pr. PP) is where the user wants to improve internal processes & is experiencing issues with their current system & process in place.
- Productivity Pain Point (PPP) is where the user wants to be more efficient with their time and are most likely wasting too much time with their current situation.
- Support Pain Point (SPP) is where the user is not receiving the appropriate support they need, especially at critical stages of the journey map.

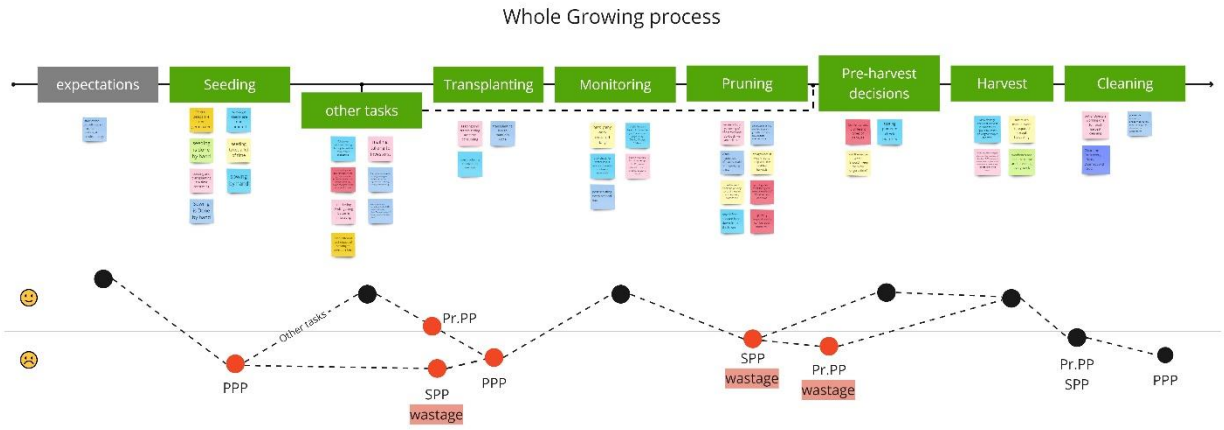



Figure 29. Phase-2 Whole Growing Process

The journey map was created in three different journeys, each specific to the tasks accomplished -Farm setup, Growing process, Post-harvest. Green AB wants to focus more on the pain points that occurred during the whole growing process, Figure 29.

**PPP** is observed mostly in the whole growing process of the journey and it occurs when there is manual work done. Tasks such as seeding, transplanting, and

maintaining the farm are where the farmer wants to save and utilize the time better.

**SPP** is observed in the critical stages of the whole growing journey like germination and pruning. Germination and pruning are the stages where assistance is needed to understand the problems as wastage is seen, fairly high. Farmers feel discouraged due to a lack of support from the industry and this eventually connects to the Process pain point.

**Pr. PP** is mostly observed in the whole growing process. Farmers are worried, they have certain expectations from the empirical knowledge they have gained. When they see abnormalities with the leaves or observe diseases with previously grown plants, they feel confused and worried, thus consult external parties or agronomists. Marketing is another area where farmers want to improve and bring in more customers.

These Pain points won't be sufficient for the study, the farm setup and post-harvest journeys will also be taken into consideration as they too are contributing factors.

**FPP** is the biggest pain point in this line of business. Recovering the initial investments of the farm is always the first preference, as farmers fear that the business may fail due to a lack of awareness among customers.

The different journeys are presented in Appendix E. The survey quantifies these pain points further.

### 4.5.1 User steps

User steps, demonstrate each stage from the journey map. The first stage is deciding what plant to grow which drives the expectations of having successful yields.

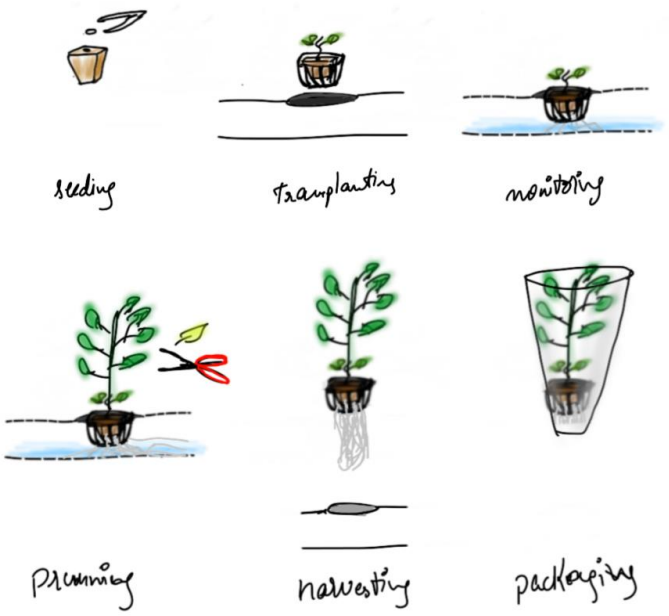



Figure 30. The user steps in the whole growing process.

The next stage would be to start the germination process. **Seeding** needs to be done in the growing medium so the seedlings can anchor themselves and, placing them in an enclosure for germination is beneficial to maintain moisture levels. The saplings are placed until cotyledons appear.

After cotyledons appear, they are **transplanted** into the system as shown in the figure, which is known as the main grow area where the plants receive light for photosynthesis.

The plants are **monitored** daily to check whether the nutrient levels are maintained or not to ensure the plants are receiving the right amount needed for growth. Parallel to this stage, other plant cycles are carried on in the farm, that can be from seeding a new plant to harvesting them or working on R&D. Management tasks are also strategically planned out in this stage.

**Pruning** is evidently the most important stage. The plants need to be checked from the roots to the leaves for diseases or abnormalities. The appearance of the plant is an important factor to every indoor farmer who is supplying to the Food & beverage industry. In addition, the interviewees also pointed out specific mold formation on the growing medium that can be harmful to humans. The volume of the plant also plays an important role here. If the plants do not meet certain visual requirements the plants are discarded, which leads to a good amount of wastage.

From baby leaves to matured leaves the customer demands the plant cycle determines the harvesting. The **Pre-harvest decision** is the strategic stage which is different for every farmer. This comes down to the experience the farmer has and decides whether to keep the plants for a bit longer or harvest them. Harvesting decisions are based on the appearance and days of the plant.

**Harvesting** is the stage where most of the wastage is seen. One of the main reasons seen here is the mishandling of the leaves. Thorough **cleaning** of the growing area is done to be ready for the next plant cycle to be grown.

### 4.5.2 Resources

A Crucial Part of indoor farms is the resources available to run the best way possible and to generate maximum benefit for the farm. It is always best to determine what resources are needed. Green AB wanted to investigate the resources needed in the Farmers' journey. Figure 31 illustrates the important resources plotted at the particular stages of the whole growing process.

Time is the biggest resource. The production time needs to be efficient. This efficiency promises consistent delivery and can also reduce man-hours and reduce labor costs.

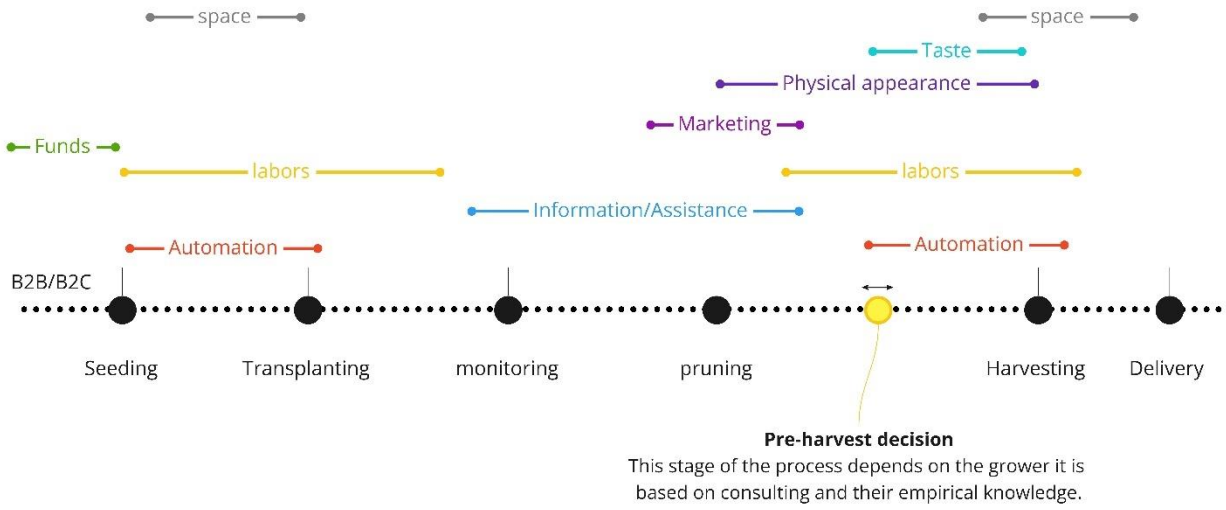



Figure 31. Resources required in the journey.

Farmers always have the need to scale up their farm, a resource such as space is needed to do R&D, expand the seeding and harvesting area to increase the yield, or build a new farm in a different city to expand the customer base.

The taste and visual appearance of the plant are crucial to check if the plant is ready for harvesting or not. The inability to use these resources will cause a process pain point and in return causes farmers to seek help from other farmers or consultants.

For small-scale and mid-scale farms, human intervention and manual labor are important during seeding, transplanting, and harvesting. Farmers express labor

as a very important resource and lacking this resource will again cause a process pain point.

Since farmers claim they do not have enough knowledge of plants, external assistance is needed to better understand the plant health and abnormalities formed on the leaves. Identifying the abnormalities and diseases will help the growing experience as per the interviewees when talking about monitoring.

Since there was a productivity pain point regarding seeding and transplanting and how they were time-consuming, farmers aim for installing automation in the farm to save time by using machinery to seed and harvest. Of course, this would need investment, and for the farmers acquiring this resource is still planned for the future.

## 4.6 Survey

A survey was created to quantify the pain points. This survey was posted in multiple Facebook groups and Subreddits centered around greenhouse farming, vertical farming, hydroponic farming, aquaponics, microgreens, and lettuce growers. To filter out the commercial farmers, different sections were created for the respondents. These sections were - Commercial indoor farmers, home growers/ Hobbyists to commercial and, home growers/Hobbyists. The responses from people who want to start commercial were used to identify initial problems of commercial indoor farming. The responses from home growers identify reasons for buying/not buying indoor-grown products and what motivated them to make that choice. It was also interesting to see 11:4 male to female ratio of commercial indoor farmers. The survey was analyzed using an Excel sheet and the responses are summarized below. For survey responses see Appendix F.

### Survey insights:

### Commercial indoor farmers - Large-scale farmers

The large-scale farmers (+1000 sq meter) have an average of 7 years of experience in indoor farming. This leads to them, gaining a lot of knowledge on plant growth and the business, in and around indoor farming.

Most of the respondents see wastage of plants occurring during the harvesting and observe an average wastage of 12.8 % in the whole plant cycle (from seeding to packaging). This wastage occurs due to various reasons such as hot weather conditions, shipping errors such as laborers damaging the pods, over-ripening, pest damage, appearances, textures, and size of the plants. Large-scale farmers have come up with a preventive measure to overproduce in order to avoid a supply shortage when it’s ready for harvesting. The survey also says people grow

the same plant over and over and it is followed throughout the year to supply the customer demands.

When it comes to the operations at the farm, water leaks and HVAC systems are the main problems of large-scale farmers. The readings are manually collected and analyzed for future harvest as this data is important for future references.

To help with the farm activities and tasks, laborers are hired who work 8-10 hours daily and these workers do not have any kind of education in agriculture or other related fields which implies that the labor point of view is not that significant and it's just everyone doing a certain task that they have been assigned on that day or week. The laborers are a very important resource as the farm very much relies on their competencies. Apart from labors, external consultants play a big role when it comes to helping with the growing and plant problems as this saves time and expenses.

Although having a background education is important, a simple crash course is also what people have attended. Most of them make their everyday growing decisions based on their experience and gut feeling. They are harvesting based on empirical knowledge, customer demand and, the appearance of the plant. They are extensively experimenting on their plants to gain knowledge for future harvests. To them as of now, increasing the value of their produce, having consistent delivery, and increasing the profits are important.

The problems encountered frequently are plant pests, molds, and tip burnings. Most farmers have a pretty decent knowledge of what might be the problem, this leads to them usually noticing an irregularity or any plant disease before it is too late for the plant. They end up losing 1.7% to 6.4% on an average of the total yield every cycle.

Growing in the summer can be very challenging for greenhouse growers due to the harsh sun. Heatwaves are problematic for the cooling capacity leading to delayed ripening and fussy flower growers. In-house experimenting is a very resource-intensive process for both mid-scale and large-scale farmers. The large-scale ones would invest in scaling up, automating the farm, and increasing efficiency in the growth cycle which they would do by hiring an expert grower. Their customers are generally direct customers, supermarkets, food markets, and restaurants. They also want to cut down on the packaging costs. Their customers need good quality, traceability of the product, and seasonal crops in their region.

The biggest concern for large-scale farmers is not being consistent in terms of, the quality that they are providing, production of the plants, and making profits.

The important factors that are holding these large-scale farmers from scaling up are funds and space availability.

## Small-scale farmers

Small-scale farmers (less than 500 sq meters) and mid-scale farmers (600-900 sq meters) have an average of 3 years of experience in indoor farming.

In the beginning, a few have a hard time finding the right contractors to help them set up the farm with components and vendors to source materials in their region.

They have observed a wastage of 12% happening while transplanting, pruning the plants, and packaging. The reasons are appearance, seeding issues, pests, electric problems, basically not everything gets picked during harvesting. The common general problems faced during maintenance are drip leaks, HVAC, nutrient deficiencies, pests, and temperature problems.

It is also seen that everybody wants to repeat the same plant. The number of laborers hired is reasonable for the farm size and they spend around 3.5 hours every day analyzing and doing the task. Most small-scale farmers believe that the laborers don't need to have a background in agriculture or other related fields.

The small-scale farmers make their everyday growing decision based on their empirical knowledge gathered from the previous harvests and are also harvesting based on the appearance of the plant and consulting external parties for that matter. To them increasing their profits and providing a consistent delivery is the priority and then to increase the value of their produce.

Problems encountered most are molds forming on the growing medium, plant insects/pests, tip burnings, and root rots. The knowledge they have gained allows them to notice irregularities and diseases in the plant cycle but that doesn't mean they have saved the plant by noticing it too early, they still want to react faster to prevent the "Disease festival". Due to disease an average of 3.2% to 8% of wastage occurs.

A few growing failures for small-scale growing are improper pest management, electrical problems, and humidity.

Though scaling up is the main thing every farmer wants, apart from that acquiring new technologies is a new want, but the main need is to understand the market demand and to increase the growth efficiency by reducing wastage. With the small-scale farmers, the focus is on supplying to direct customers and a few are interested to supply restaurants and supermarkets.

The factor that concerns small-scale farmers is not being consistent in terms of production and productivity.

## Home growers/hobbyists to commercial growers

People who want to start commercially are validating the model and most have full-time jobs. The respondents have started this as a hobby and have no educational background in agriculture or other related fields. It is common to agree that everyone starts with entry-level plants in this industry and aims for more energy-intensive plants. People are struggling with the nutrient solution intake of the plants, lighting, the components of the system, and maintaining the temperature of the environment.

Factors holding back from going commercial are investments, entrepreneurial experience, space, knowledge, and growing experience on a larger scale. They have a pretty good domestic knowledge which might not be that useful in the business as customer demands have to be met. People feel starting an indoor farm is risky. Sure, they would be happy to have some form of technological support in the beginning but have to invest accordingly and that is where the issue lies.

The important aspects to start commercial full time are hiring the right personnel, finding a steady customer base and supply to that demand, reducing CAPEX in terms of energy usage, and most of all, support from the locals and government.

## Home growers/hobbyists

They are less familiar with indoor farming before they are satisfied with growing in their homes or personal spaces. They have no further plans on growing commercially. People who do not buy products from an indoor farm think it's better to grow their food and it's cheaper this way. People find indoor growing to be a beneficial business in terms of the potential it can achieve in the long run. They find a sense of optimism in the near future.

# 5 Insights

### Specific Growing problems

The main concentration was on the growing problems, and these are drawn from all the interviews, surveys, farm visits.

Although farmers have got a hold on growing entry-level plants like leafy greens and microgreens which do not consume a lot of energy and resources in the process, the main goal is to supply the demands.

1 Aquaponic growers are struggling to maintain the salinity of the fish tank. Problems with pH because baking soda is used to raise the pH of the fish water and this doesn't suit the plants.

2 Seeding the plants is done by hand, it takes time and effort to seed properly into the growing medium. Not all the seeds germinate which is considered a significant wastage of resources. Different types of seeding techniques are tested out to bring the best output and 7 out of 10 farmers fail to achieve 100% germination. This reduces the production efficiency of that particular batch of plants.

3 After transplanting the plants to the growing area, the plants are monitored every day and here it gets tricky for farmers. The process needs to be taken care of in such a way that early diseases or abnormalities have to be dealt with in the early stages of the cycle and this is made easier with the help of technology (IoT devices).

   a. People grow in their way of understanding as most of them use Rockwool, some use peat as the growing medium in their NFT systems. The main problem is the confusion about maintaining water levels even after several harvests. The gutters are flooded and lead to the overflowing of water.

   b. People chose to grow different plants in the same system which makes it difficult to maintain. The nutrient recipe is different for every plant and requires a different rate of water distribution. This water distribution rate is challenging.

   c. Using the correct system for growing vegetables is crucial. Although NFT is the best and a simple method it does not work for resource-intensive vegetables. The roots block the channel and cause a shortage of water reaching all other plant roots in the system, this is often observed in newcomers.

4 Grow lights play an important role as the intensity of the lights is tweaked based on the expected yield. This creates heat in the farm which then leads to

HVAC and air circulations problems, so, this could be a risk factor growing in tropical climates. This problem is generally observed in container farms and greenhouses, where firsthand, tackling the heat will be the biggest issue. The plant placements at the end of the growing systems, observed in the interviews, did not receive light resulting in long stem structures and latex present in them.

5 Monitoring the plants is a daily job and covers most of the tasks in the plant cycle. Pruning is essential to prevent the risk of diseases from spreading to other leaves. And this needs a lot of patience and effort if done with the help of manual labor. It is exhausting to check for irregularities and diseases with the fact of managing such a large area of the farm.

   a. Wastage is observed here and it is a fact to consider because the pruning of a lot of plants can be related to a lot of irregularities formed due to nutrient deficiencies, electric problems, and pest damage. The farms are focused on delivering high-value produce and fail to maintain the consistency of successful yield with an average wastage of 1.7% - 6.4% (large-scale) and 3.2% - 8% (Small-scale).

   b. Checking for root rots is a hassle as every root must be inspected. The farms with NFT and vertical tower systems have difficulty in this area.

6 Nutrient deficiencies were seen as a big issue while growing. Deficiencies like iron, nitrogen, potassium, phosphorus, calcium nitrate and other microelements added to nutrients.

7 People have experienced a lot of pest damage, root rots, tip burnings and, harmful mold formations on the Growing medium.

8 The main need for every indoor farmer is to have a preventive measure to certain problems occurring on a day-to-day basis (mostly plant health, system malfunctioning, and hiring knowledgeable personnel). The main requirement for every indoor farmer is also biosecurity and due to the technological advancements, Large-scale farmers want the indoor farm to sustain without human interference and very few have reached a certain level. When it comes to the small-scale and mid-scale farmers this is not the case, surely, they want to reach a similar technological level but it is important for them to have human intervention as they are very much reliant on manpower to undergo tasks at the farm.

9 The pre-harvest decision is a mainly subjective decision for every farmer. From supplying baby leaves to mature leaves, everything depends on the customer’s demands.

10 Harvesting has specific problems for certain farmers who have faced male employees that had crushed the leaves when harvesting, eventually destroying the product. It was interesting to see a gender-specific social norm from a female interviewee:

*“Everyday work is more woman-suitable type of work rather than men, Females put in a lot of emphasis on aesthetics and being gentle with the plants (harvesting, seeding)”*

## Key insights

### 1 Base setup

People start with a small scale to test out the viability of the system. It is found that farmers start with the least expensive/ reasonable setup which does not require many technological advancements but the upgrades are available to farmers with large-scale investments. One main aspect observed here is that small-scale farms have not been set up in favor of the plants, but easily resource able components and growing materials for the system. The reliability of the contractor is of utmost importance for small-scale and upcoming farms.

*“I had a terrible time with my contractor.”*

*“A contractor from Finland delivered the materials for me and I had to set it myself. It took me months to finish it!”*

*“Trusted name is important in buying tech”*

### 2 Well-being

Indoor farms are therapeutic in a way people can benefit from it; this adds to a highly effective social value brought into the urban environment.

*"It’s soothing and meditative"*

*"The greenhouse is my therapy; I am relaxed when I leave"*

### 3 It’s not just about growing

It becomes an entrepreneurial task to look after the marketing, logistics, and future plans of the farm. Customers buy regularly from the farmers market and it has become the biggest competition in cities. Surviving on phase 1 crops (Leafy greens) to break even is debatable, as a large portion of customer demands are also inclined towards phase 2 & 3 crops (vegetables, fruits, and staple crops). Farmers concentrate more on marketing to introduce their produce and also to retain occasional customers. Balancing between growing and managing these tasks is challenging.

*“I have to constantly educate people about my produce!”*

*“Educating the customer on the value of hydroponics is a constant battle”*

*“You may have a good product, but who's going to buy your product and for how much someone is going to buy it for”*

## 4 R&D

In-house experimentation is agreed to be a resource-intensive process. Providers, only collect plant data, experiment on nutrient recipes, develop R&D models, home grower kits, hydroponic greenhouses. Pure-Growers on the other hand are purely focused on consistent production and understanding the growth of variety of plants.

Market demands vary from country to country and cause an effect on growing habits. Farmers are in desperate need of empirical data. Most are happy with their knowledge on growing and what they have achieved; having that said it isn’t recovering the initial cost invested, at least for pure-growers. To achieve that, the focus is on diversifying indoor-grown plants. This is the main reason for doing R&D. Growing different plants is not an area many have achieved success.

“*The farm data helped me optimize my vertical tower product*”

*“I did not grow tomatoes because it didn’t provide me scientific data”*

## 5 On Repeat

Many farmers, all-scale, are keen on repeatedly growing the same plant. It is observed here that NFT and Deepwater cultivation are the most used growing techniques for growing leafy greens and microgreens as they are easy to grow in. Having a sense of safety in repeating, and a dependable plant that can be sold any time of the year is seen as significant.

*"I’m growing only lettuce and microgreens all year and it’s better this way!”*

*“We are following the same nutrients for the herbs”*

## 6 Data is all I have

Empirical knowledge is gained by growing and testing different methods, plants, hacks, etc. All the produce that is grown now is mostly beginner plants or phase 1 plants and there is a need to step further to new plant phases, 2&3, which are essential for the market demands.

This data, mainly plant recipes, is what all the indoor farmers, both the providers and pure-growers have, and it’s nothing but the data and experience gathered from previous harvests. Farmers are not willing to share certain methods of the plant cycle (Germination, Pre-harvesting) as they want to stay competitive in the

business. Data metrics are valuable, whether the data is collected manually or through IoT devices, data is data**.**

*"Data is all I have to base my decisions on"*

*"Switching to other plants makes my life more interesting and it also provides much more scientific data to forward it to my customers"*

### 7 Consultants

Farmers are lacking in terms of research materials which creates a knowledge gap. As a countermeasure to resolve any problem that occurs during growing, they consult external parties or agronomists. Newcomers and small-scale farmers feel the indoor farming industry is not supportive in terms of helping with their growing process so they consult for a reliable source of information. This drives up the cost. Empirical knowledge is important, having assistance in growing is of equal importance as it helps habituate, farming as a lifestyle.

"*The reality is much more different, once you get into the business, it's a different ball game*"

*"The key is to understand diseases, what is happening with the plants"*

### 8 What is the value?

Farmers, perceive the produce, to have a high product value and harvest based on the visual appearance of the plant. They see the worth in spending more, whereas the customers consider indoor-grown produce to be 'expensive' and choose not to buy them. Although price and value are not in relation to each other, not many see the worth in buying these products, causing the decline in the value. The indoor food "communities" are very few compared to people who choose traditionally farmed plants. These communities are loyal customers, along with a few occasional buyers that are helping the farms generate income. There is lack of awareness among the customers.

*"It's just a mental block people have"*

*"The most important problem will be to get a high price for the products."*

### 9 The potential customers

It is seen that the majority of the indoor farmer are focused on supplying only to the Food and Beverage industry like restaurants, food markets, supermarkets, direct customers, etc. A handful is keen on selling it to other industries where the value and the selling price of the products fit the farmers' financial needs. Leafy greens are not only good for food consumption but can also be used in different industries which require high-quality plants. Indoor grown products are marketed as "healthy", "nutritious" and so, most of which aren't new to the customers and don't change the habits that will bring a change.

The money invested is also the main reason that is stopping the farmers from quitting. Indoor farmers have no choice but to sell the crops as “premium”, considering the investment and the resources involved to make this technology viable. Due to this fear, some indoor farmers are selling the plants at a “throwaway price”.

*“Weekly deliveries to the restaurants”*

*“From last year we increased to 20% subscription rates of direct customers and supermarkets”*

## Key insights revised

After the Key insights were identified from the user studies the opportunity spaces were analyzed. The growing problems identified from the user studies say that the problems are specific and need individual attention. Key insights such as, **it’s not just about growing**, **on repeat** and, **the potential customers**, were the areas which can be only looked at by the Farmers. The remaining key insights focused more on the growing problems and the root cause for growing problems, so they were concentrated as they create room for opportunities to work with.

The key insights that were concentrated are,

**Base setup**- It is learnt that small-scale farmers do not have access to reliable vendors which restricts them to try and test different growing materials. This leads to increased operational costs. Not having proper growing materials can be the root cause of the growing problems that can occur in the later stages of the plant cycle
(as seen in the interviews).

*“I had a terrible time with my contractor.”*

The key insights also presented a pattern with **R&D**, **Data is all I have**, **Consulting**. R&D is basically carried out as there is no research material available to refer to. The trials and testing of growing diverse plants allow the farmers to gather data on their harvests and growth cycles. It has not been very much successful as they have observed uneven growth and experienced wastage because of the Knowledge gap. They consult external parties and agronomists to understand the plant growth to fill the knowledge gap which again increases operational costs. An opportunity space is to bridge the gap between empirical knowledge and the farmers.

*"Your product is never 100% right"*

*“Data is all I have to base my decisions on”*

“*The reality is much more different, once you get into the business, it’s a different ball game*”

*“The key is to understand what is happening with the plants”*

**What is the value?** – A major problem with the farmers was to translate the value of their produce to the customers to acquire new customers. Though the problem was outside of the growing problem scope it was interesting to see what could be done with this area of opportunity.

*“It’s just a mental block people have”*

*“The most important problem will be to get a high price for the products.”*

# 6 Ideation

The Key insights helped to generate a few how might we questions. These questions allowed the research to ideate. Before the ideation stage, a set of 'How might we' questions were created in order to turn user problems into an opportunity by asking 'How might we?'.

- How might we ensure newcomers and small-scale indoor farmers have access to reliable vendors and contractors?
- How might we help indoor farmers receive support and provide empirical knowledge to solve their growing problems?
- How might we spike the interest in indoor farming to urban dwellers.

The ideation generated two concepts, presented below

Concept #1- Community-based app

The idea is a community-based app that can be used by farmers from different countries and build a forum wherein they share their experiences, learn from one another, and teach upcoming farmers about the business.

The newcomers can use the app, on a subscription basis, to contact other experienced farmers and receive help to solve the typical growing problems that occur. One of the pain points was the reliability of vendors, this service gives indoor farmers exposure to reliable and viable vendors to source materials.

The user studies show that there is a lack of knowledge observed in tackling growing problems and growing different plants. The app consists of a feature wherein the users can receive tips generated from other indoor farmers, agronomists, and university researchers so they can have access to reliable information to help improve their knowledge.

Main aspects of the service

- Genuine and verified profiles of Commercial indoor farmers.
- Multiple forums within the app to specific topics.
- Contacts of vendors, provided by experienced indoor farmers.
- The user chooses the plant that needs to be grown and tips are provided. They can further choose the information that interests them.
- The users can share videos and photos to ask, and provide clear information within the platform.
- Advertisement posts by providers.
- Regular updates about the business.

Concept #2 - Farm therapy

The concept of farm therapy was to build the value of the farm to the urban dwellers.

The concept is derived from the key insights and contextual interviews that involve positive experiences about the farm environment. This environment could help tackle mental health issues such as anger, anxiety, and stress.

'Well-being' + 'It's 9 to 5' + 'what is the value', are the combination of key insights used.

The concept consists of a service that provides farm tours and utilizes simple farming tasks as therapy, to the urban public. The service allows not-so-complicated, farming tasks such as seeding, transplanting, and pruning, as these tasks are generally carried out manually. The aim of the service is also to educate people about indoor farming and bring a highly effective social value into the urban environment.

The main aspects for the service would be:

- The urban public can book a farm tour or a therapy session through farm websites and social media platforms.
- Opportunity to help in the farm chores contributing to different tasks.
- Free samples of the produce would be provided to the public.

Experiential selling was the inspiration behind concepts to answer the 'How might we' questions.

## 6.1 Tigger materials:

Concept #1 – A Paper prototype, was made for the user to test the feature of the community app and receive feedback.

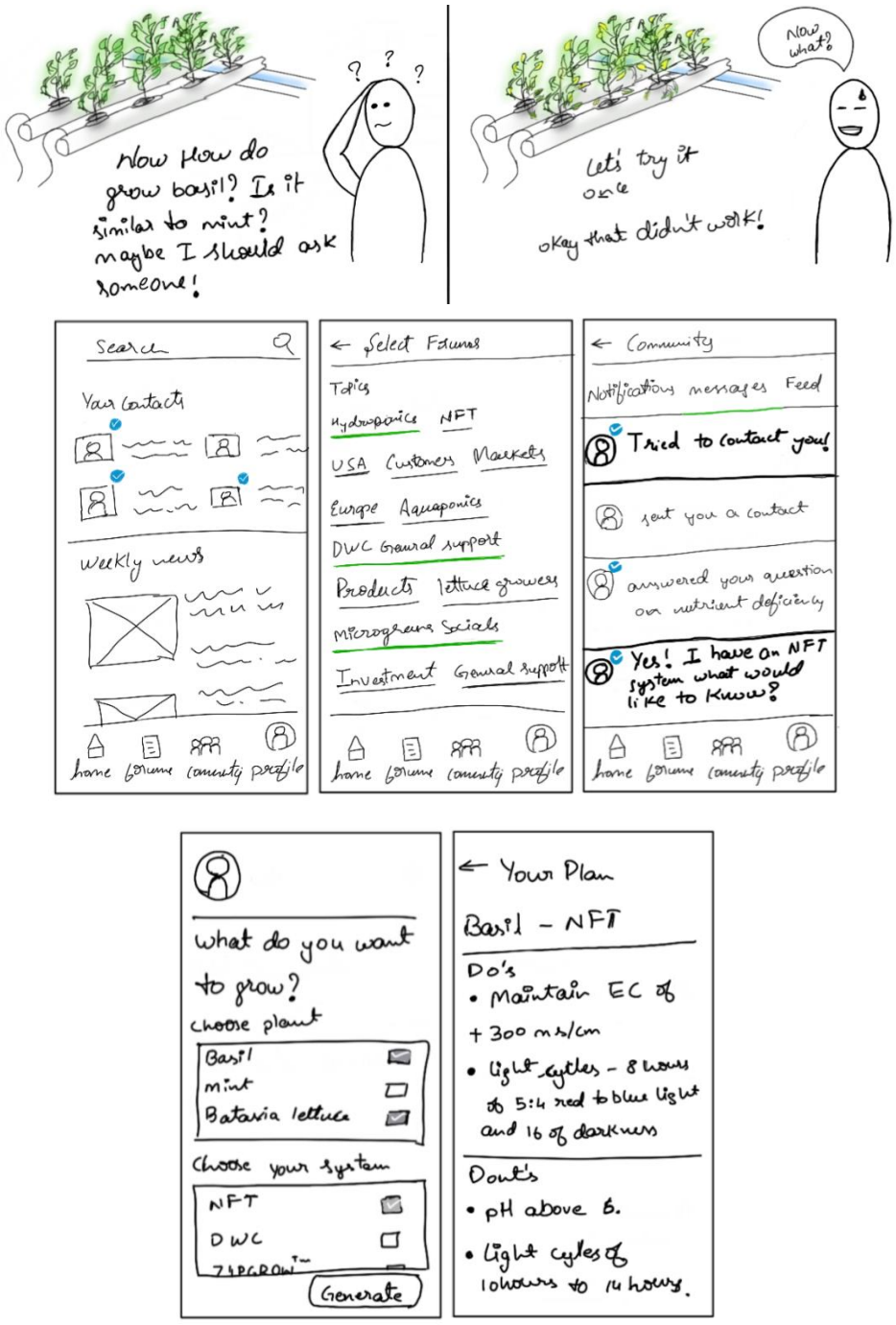



Figure 32. Community app trigger material.

Concept #2- Trigger material for the user to visualize how the concept would work and how to educate the customer on the value of an indoor farm.

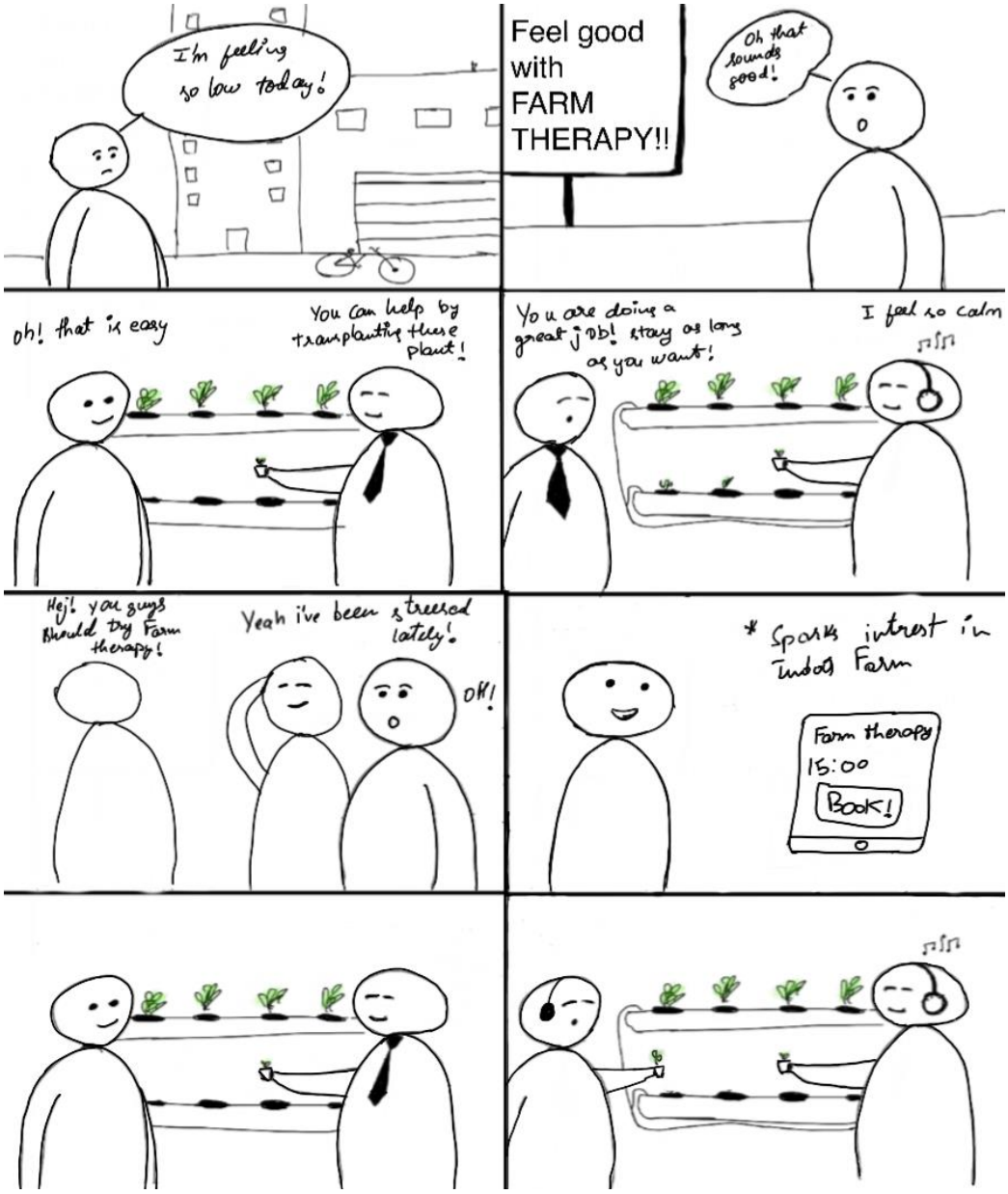



Figure 33. Farm therapy trigger material.

## 6.2 User tests

The most important inspiration in the ideation stage was the idea of experiential selling. Incorporating this tactic into the ideas was the key to solving most of the growing problems. The two concepts were tested on one Small-scale and one mid-scale farmer using low-fidelity wireframes. The low fidelity wireframes used for the user test are presented in Appendix G. This test was conducted to check the validity and desirability of the concepts. The insights were as such:

**Community app** – The concept was appreciated as it is useful in finding vendors during farm setup or during expansion. The users found the idea of contacting

other indoor farmers for finding solutions, helpful. This app is beneficial for newcomers and small-scale farmers and is considered of having high potential.

Although this idea has a reliable way of providing information to the farmers, it does not necessarily mean the growing problems could be solved right away because of the connectivity issues between the farmers.

**Farm therapy** - The idea of farm tours was not new, but using it to improve health benefits was appreciated. There was a concern regarding human contamination and this would risk the plants' health. This concept would be possible only by taking precautionary measures like imposing non-negotiable protective wear on the people to keep the plants safe.

Learnings from the user tests: The tests found both the concepts to be desirable (users want it) but cannot be validated based on the number of user tests. It is safe to say that the community app would be helpful but needs quantification and the small-scale and mid-scale farmers expected the ideas to solve the growing problems as soon as possible.

# 7 Discussions

This section will briefly cover unexpected findings before research questions and results are discussed. The Human-centered approach was followed thoroughly during the user studies and as a result, the outcomes were possible. Although the aim of the project was user research, the services from ideation, are very much in line with the purpose of the research.

During the user studies, the pain point of not having reliable vendors or contractors was unexpected findings from the in-depth interviews. As harvesting decisions relied on customer demands it also relied on certain visual conditions and the taste of the plants. These are a few findings that the author believes have possible significance when it comes to the decisions of an indoor farmer and weren't expected in the user studies.

## 7.1 Research questions

**What are the day-to-day activities of the indoor farmers in the whole growing process?**

The methodological triangulation allowed us to understand daily practices. To put it in simple terms the day-to-day activities are to check the plant health, nutrient levels, water/air temperature, and maintain the balance every day.

**How are they affecting the indoor farmers in the whole growing process?**

Checking the nutrient levels and water/air temperature on a daily basis makes farmers' life easier as they are updated on the farm and the plant cycles. Keeping the farm/greenhouse up to date helps build trust in the empirical knowledge they are gathering and also build authority over the growing process. The daily monitoring helps them keep an eye on plants and remove non-uniform plants. Achieving success in practicing these daily activities project a certain level of confidence and this definitely contributes to bringing a positive impact in the process.

Apart from the positives, they gain a negative experience as they do not have any assistance when there is a drop or rise in the nutrient levels and temperatures. This results in pooling resources to experiment with the plant cycles just to be certain of what the next cycle should be checked or maintained regularly. Apart from the daily monitoring, there are other tasks that have to be done during the day depending on the particular stage of the plant cycle

**What are the challenges and pain points faced by farmers?**

From the business point of view, the farmers face entrepreneurial challenges such as marketing, educating customers, customer acquisition, and sales, etc. The challenges during the growing process were, not having a decent knowledge of growing new plants, unsuccessful germinations, understanding plant health, and difficulty maintaining less plant wastage.

The pain points were categorized into 4 types, financial pain point (FPP), process pain point (Pr. PP), productivity pain point (PPP), and support pain point (SPP). Out of these pain points, FPP was the biggest pain point as ROI was important for farmers. SPP was experienced during the pruning and germination process resulting in the Pr. PP was mostly during the pruning stage of the growing process. PPP was observed as seeding the transplanting were time-consuming for the farmers. The qualitative study also understands that farmers have a need for knowledge and the change in customer's value will bring profit to the business.

**How are the challenges and pain points, different for small and large-scale farmers?**

Every indoor farmer goes through different problems in the journey, there are certain differences observed in the Growing process as well. The biggest challenge faced by small-scale farms is diseases and pest damages, affecting production efficiencies. The large-scale farms do notice diseases but have resources such as employees and research information, that helps them mitigate the plant damage. Access to resources is the main difference between small-scale and large-scale farms.

**How can suggested solutions help the farmers' needs?**

The opportunity space is challenging because every farm is different in terms of the system it is using and the growing problems caused. It's challenging to find a collective solution because what seems to be a challenge for one farmer, may not be a challenge for another farmer and so. The community app acts as a collective solution that helps the farmer achieve their functional and emotional needs from the elements of value pyramid, with networking and reaching out to other indoor farmers, farmers can share videos and photos on the platform which exposes them to the empirical knowledge gained by other farmers, it helps by providing them with information about reliable vendors, and gives access to experienced indoor farmers to learn more about the business. The farm therapy concept helps farmers in raising awareness and educating urban dwellers about the benefits of an indoor farm in urban areas.

Farmers are not very keen on sharing the data they have gathered but are willing to give advice to other farmers, hinting, how the solutions should be created in the future.

### Adoption of technology

It is clear that every indoor farmer wants to reach a certain point in their business where there is less human intervention in the process. To reach this point there need to be initial investments as well as operation costs to run the farm. The pure-growers are skeptical about buying or outsourcing technology from providers or companies. They believe it is risky and want to make sure the technology is viable. Most of the farmers aren't much interested in outsourcing technology but would buy if the technology is cost-effective, has good reviews from other farms, and most important of all, help them stay competitive in the business. The farmers are keen on technology that can manage the farm/greenhouse and utilize the resources effectively.

## 7.2 Results

The result of the project is the Qualitative analysis that explores the farmers' day-to-day activities in the Farm/Greenhouse suggest they have information but lack knowledge. The three different Journey maps, Farm setup, Whole growing process, post-harvest, helped to identify the pain points. The survey quantified these pain points and gave a better understanding of commercial indoor farmers, people who are starting commercial indoor farming, and home growers. The insights developed from the project are applicable to the existing market in the year 2021. As the project progressed into ideation, two concepts were developed, the community app and the farm therapy. These concepts were most applicable to the smaller-scale farmers.

## 7.3 Future work

The project resulted in two concepts, each addressing a different challenge. Due to time constraints, the concepts were not quantitatively evaluated. All though these concepts are services, it does not mean they are the final solutions in helping the farmers. The concepts suggested in this project should be evaluated quantitatively to check their validity. If the concepts are not validated by the users, then the primary focus should be shifted to creating solutions that solve the farmers growing challenges in a short time span after they are discovered. In order to improve the concepts further, an ideation workshop with the indoor farmers would be favorable.

The questionnaire, survey, and insights can also be used for continuing further research by expanding or diversifying the users to focus on beyond the current target audience. Another opportunity is to concentrate on the countries that import fruits and vegetables (phase 2 plants). Helping these countries with indoor farming can be advantageous and also contribute to reducing carbon footprint.

# 8 Conclusion

The project was able to address all the aspects of the thesis proposal and answer the research questions formed at the beginning of the thesis. The HCD framework considered the farmers' perspective throughout the design process. With the help of methodological triangulation, the exploration of the farmers' day-to-day activities and pain points it is seen how the developing indoor farming industry and its challenges affect the indoor farmers. The project concludes that the room for action in regards to collectively solving the pain points was extremely challenging. The two concepts suggested in this project, catered to the newcomers and small-scale indoor farmers, proved desirable and thus need to be validated, but Green AB can use the results to explore further opportunities.

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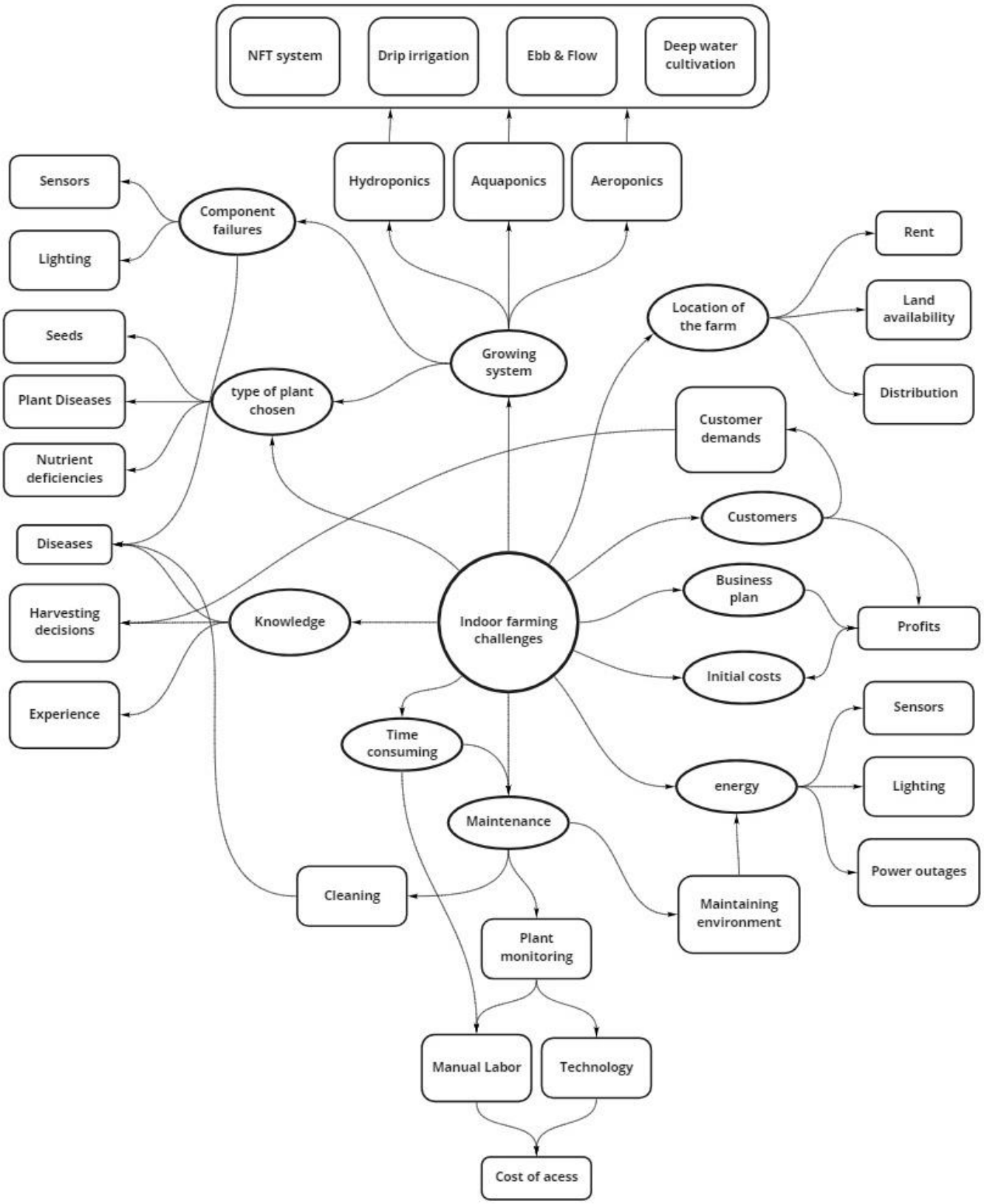
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**Webinars and podcasts**

31. How to Get Started in Vertical Farming and Urban Agriculture – the Next Big Thing or Cities. Hosted by David Thorpe, Special Consultant to Sustainable Cities Collective. http://sustainablecitiescollective.com/davidthorpe/1074146/webinar-roundup-how-get-startedvertical-farming-and-urban-agriculture-next-big
32. Urban Agriculture Audio/Podcasts. Hosted by Vincent and Dickson, Professors at Columbia University Medical Center. www.urbanag.ws Columbia University Mailman School of Public Health Department of Environmental Health Sciences.

# Appendix A – Mind Map



# Appendix B - Customer demands

## Survey Summary

- We surveyed 150 Potential Customers in Jaipur to get an idea of the current market demand for exotic greens, fruits & other vegetables by asking them questions on the following:

-Consumption Patterns for exotic fruits & vegetables

-Where do they usually source these exotic fruits & vegetables from?

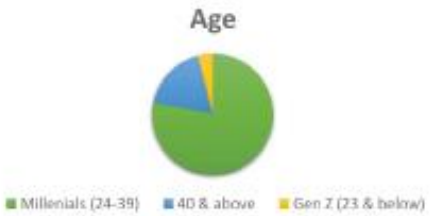
-Awareness and willingness to try hydroponically grown produce

-Their Opinion on nutritional value of current produce, preference for locally and sustainably grown produce

## Respondent Demographics

(By Age and By Gender)

- **78%** of the people surveyed were **millennials** (between the age group of 24-39 years)
- **18%** were above the age of 40
- **4%** were Gen Z (age 23 and below)

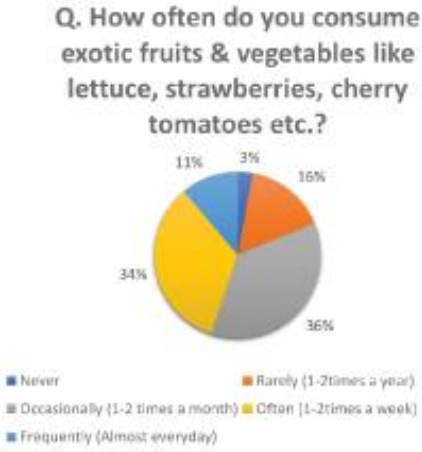



- **57%** were men
- **43%** were women




## Consumption Patterns

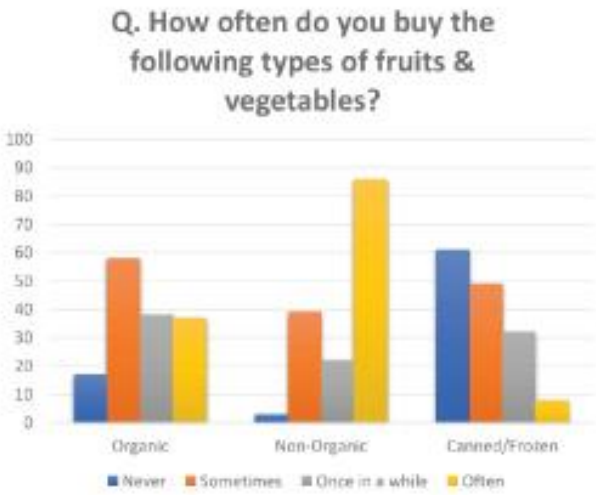
- 70% of the respondents are occasional consumers that eat exotic fruits & vegetables a maximum of twice a week
- Such low demand for exotic produce can be due to lack of availability, high price and personal preferences



# Preference for Produce Type

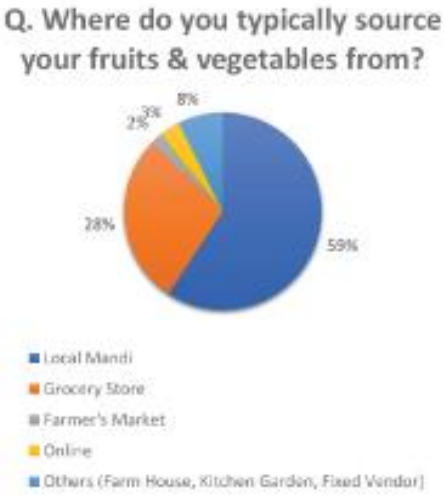
(Organic/Inorganic/Frozen)

- Majority of our respondents prefer Non-Organic produce, with 85 of them claiming that they buy it 'Often' as compared to 58 people who claim to buy Organic produce only 'Sometimes'
- Jaipur has still not fully accepted Organic produce in their daily lifestyle, but this is a trend we see going upwards in the coming years



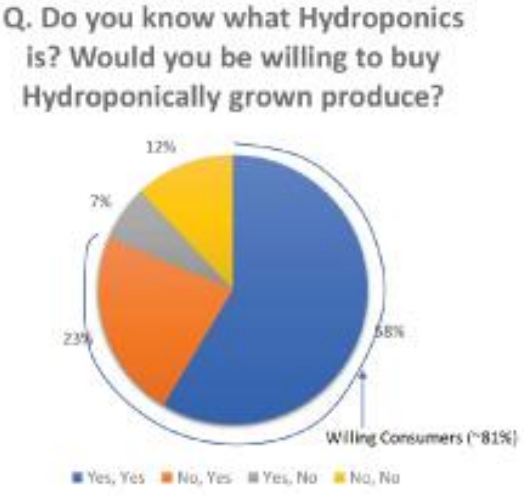

# Following the Consumer

- Mandi is clearly the most preferred option for our respondents with 59% choosing to buy their produce from there with only 3% ordering it online
- The combined 30% market of Grocery Stores and Farmer's Markets is the target audience for us as they seem to be more concerned about quality of their produce and are willing to pay for its value



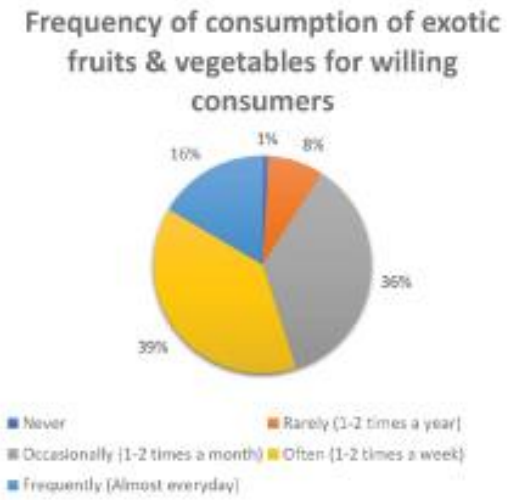

# Awareness and Willingness to try Hydroponics

- Approximately 2/3rd of our respondents know about Hydroponics, out of which ~90% show a willingness to buy the same
- This is a positive that shows us creating awareness about Hydroponics is paramount, as we can see that an aware consumer has a very high chance of conversion



# Consumption Pattern for Willing Consumers

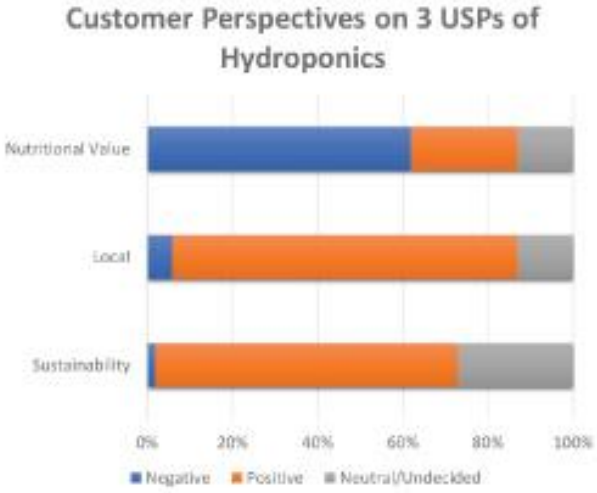
- As compared to the consumption pattern of all our respondents, the willing consumers have a similar consumption pattern with a marginal increase of 10% in frequent users




# Customer Perspective

(Nutritional Value of currently available produce/Likelihood to purchase Locally grown Produce/Importance of Sustainability in Agriculture)

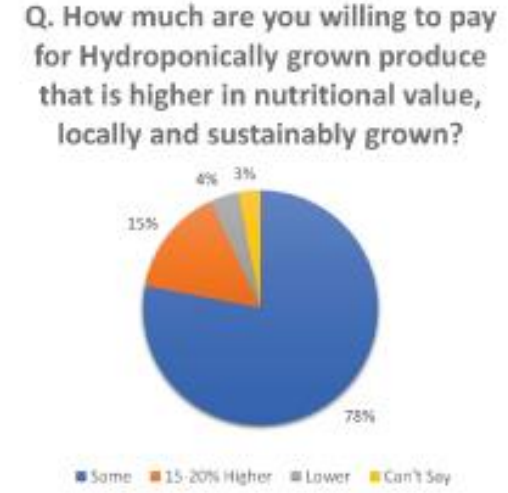
- 62% of our respondents do not think that the produce currently available in the market has good nutritional value
- 81% would prefer buying locally grown produce and 71% would prefer a sustainable farming alternative
- The 3 USPs of Hydroponics seem to play a major role in the respondents' buying decisions




# Customer Perspective

(Willingness to pay for Hydroponically grown produce viz-a-viz current options available in the market

- Even with nutritious, locally and sustainably grown produce being of high importance to our respondents only 15% of them are willing to pay a marginally higher price for it



# Conclusions

- Demand for personal consumption of exotic fruits & vegetables is still fairly occasional however, with increased availability and awareness, we expect it to rise overtime along with an increase in demand for cleaner produce
- People still prefer to purchase their produce from the Mandi with a very low online demand. Strategic offline partnerships will be important to reach the consumer
- Probability of someone trying Hydroponically grown produce is very high when they are aware about the technology, this still doesn't tackle the problem of occasional demand
- Consumers do value higher nutritional value, local and sustainable cultivation practices but are not willing to pay higher than market price for it

# Appendix C - Interview guide

Hello, my name is Sanketh Yeli. I am from India and I'm a student at KTH Royal Institute of Technology, Stockholm, studying my master's in Integrated product design. I'm currently working on my thesis project with a focus on User Experience of in-door farmers. I'm researching my target group which is commercial In-door Farmers and trying to understand their daily activities and challenges they are facing for my research. This interview will be conducted for 40 minutes and you can choose to not answer any of the questions that seem inappropriate to you.

**Background**

What is your name?

How old are you?

Where do you live?

Can I record this interview?

**General**

What is the size of your current farm? Any reasons?

What made you want to get into the farming business?

How long have you been farming? do you like it?

What's the importance of farming in your life?

Did you have a farm before this farm?

Why did you change?

How far is the farm from where you currently live?

What different aspects did you look into before choosing the location of this farm? Which of them made the most impact?

How close is the location of the farm to your customers? specific reason?

**Growing**

What type of growing system are you using? Why did you choose this?

What kind of plants/produce do you grow now? Why?

How many types of plants have you grown before?

Okay, so how does growing look like in different seasons of the year?

How do you know what to grow and when? What factors affect your decision?

How much risk do you think is involved?

Are you satisfied with your current setup? Would you have done it in a different way? In what way would that help?

What are your thoughts on wastage occurring during the growing processes?

What part of the growing stages is waste occurring?

What are the causes for this wastage?

What type of seeds do you use for cultivation (GMOs or Organic)? Why?

How likely are you to experiment with your plants during the growing process?

What are you obtaining from this experiment?

Will you go against the system/data to experiment with the produce? when would you do it and why?

**Day-to-day**

Can you explain how your work on the farm looks like

Could you describe a day at the farm? the main tasks you have when you enter, up until you leave for the day.

Can you describe how do you feel when you start the day at the farm?

When you are at the farm?

When you leave for the day?

Where do you want to spend most of the time at the farm?

Is there anything you would like to spend more time doing?

What is the average amount of hours you spend at the farm?

How much manual work is involved in your day-to-day activities? Does it vary with days? Why?

**Operations**

How many people are working on the farm and who are they (what are they working as)?

How many hours do they work in general per day?

What competencies do they have?

How is their work helping you?

What competencies are still required?

According to you, how much area can a single person manage to cover on the farm?

Do you think your farm operations meet the efficiency you wanted to achieve? Why?

Can you describe how your post-harvest process looks like?

If you were to get extra funding what improvements to your farm/way of farming would you start with?

If I say 'full control' what does it look like to you? Where does it seem important in the process?

**Mental & physical well-being**

How do you feel about the farm in general?

What do you like the most?

What do you not like so much?

How do you feel when you have started a specific growth cycle?

**Knowledge**

Have you any prior knowledge before you started farming?

Where do you get your growing knowledge from? How much do you rely on it?

What is it that makes you aware of the quality of the produce after you harvest?

How and what do you base your decisions on? (Example: after transplanting)

How do you feel after making a decision? why is that so?

How informed are your customers when it comes to the produce they buy? And how do you a line your strategy with their idea of value?

**Adoption of innovation**

How do you feel about human involvement in your farming process?

Do you think eliminating, human involvement from the growing process is important for you? why do you think so?

What resources do you think can help you achieve it? Why would they help?

If I were to say you would outsource technology, where would you outsource it from?

Since this business is full of new innovations emerging every day! What are your thoughts on adopting these new innovations?

What is most challenging in adopting and implementing new technology?

Wouldn't it better if you could just 'Buy' these technologies from another farm? How do you feel about it?

**Challenges**

What was the biggest challenge(es) with setting up your farm?

Can you name some common/general problems you are facing during the maintenance of the farm?

Any problems/difficulty you faced that has caused the farm some major disturbances?

How much attention did you spend on this problem?

Is there anything you would like to change to improve this challenge/experience?

How and what do you base your decision on? (Give an example - harvesting time)

What resources do you think are important for your day-to-day activities at the farm?

**Customers**

Does your customer have any particular demands that you are faced with?

(For example, delivery times, seedling, etc.)

What do you think is most important to your customers?

What do you include in your fresh sheets? Is it possible to see one?

If you are not comfortable, can I at least get an idea of how you prepare it and how often do you update it?

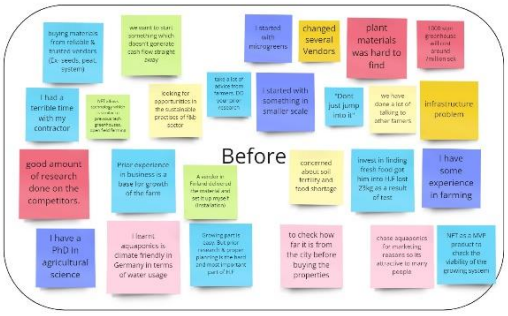
How convenient is the location for your produce to reach your customers?

How do you place your farm in relation to your competitors?

Do you see any particular threats?

Do you keep up with all the innovations in indoor farming? Anything you would have opt for? Why?

# Appendix D – Clustering



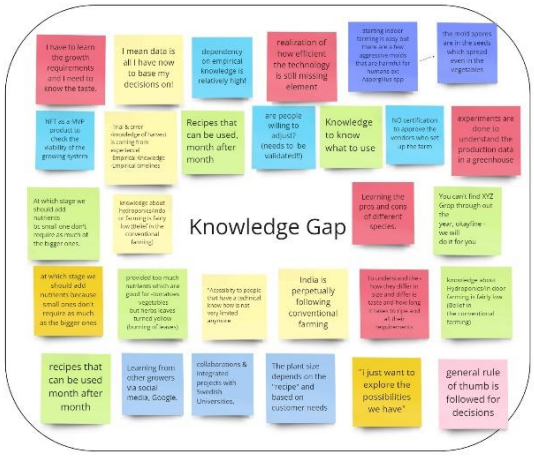




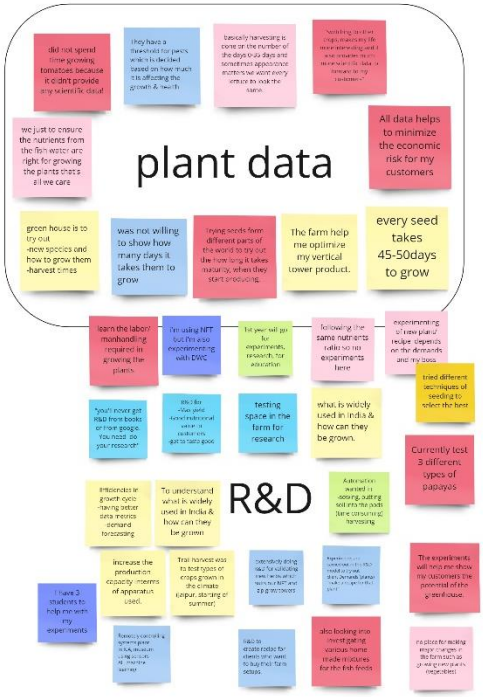


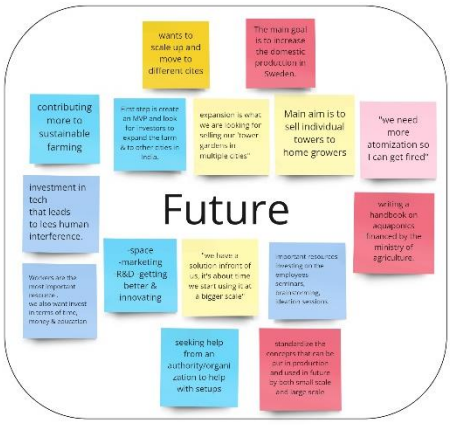




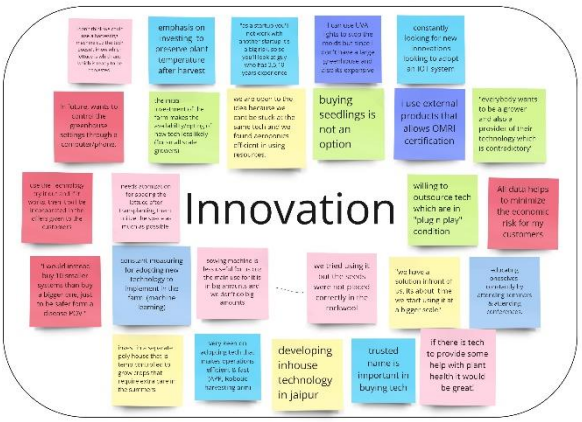


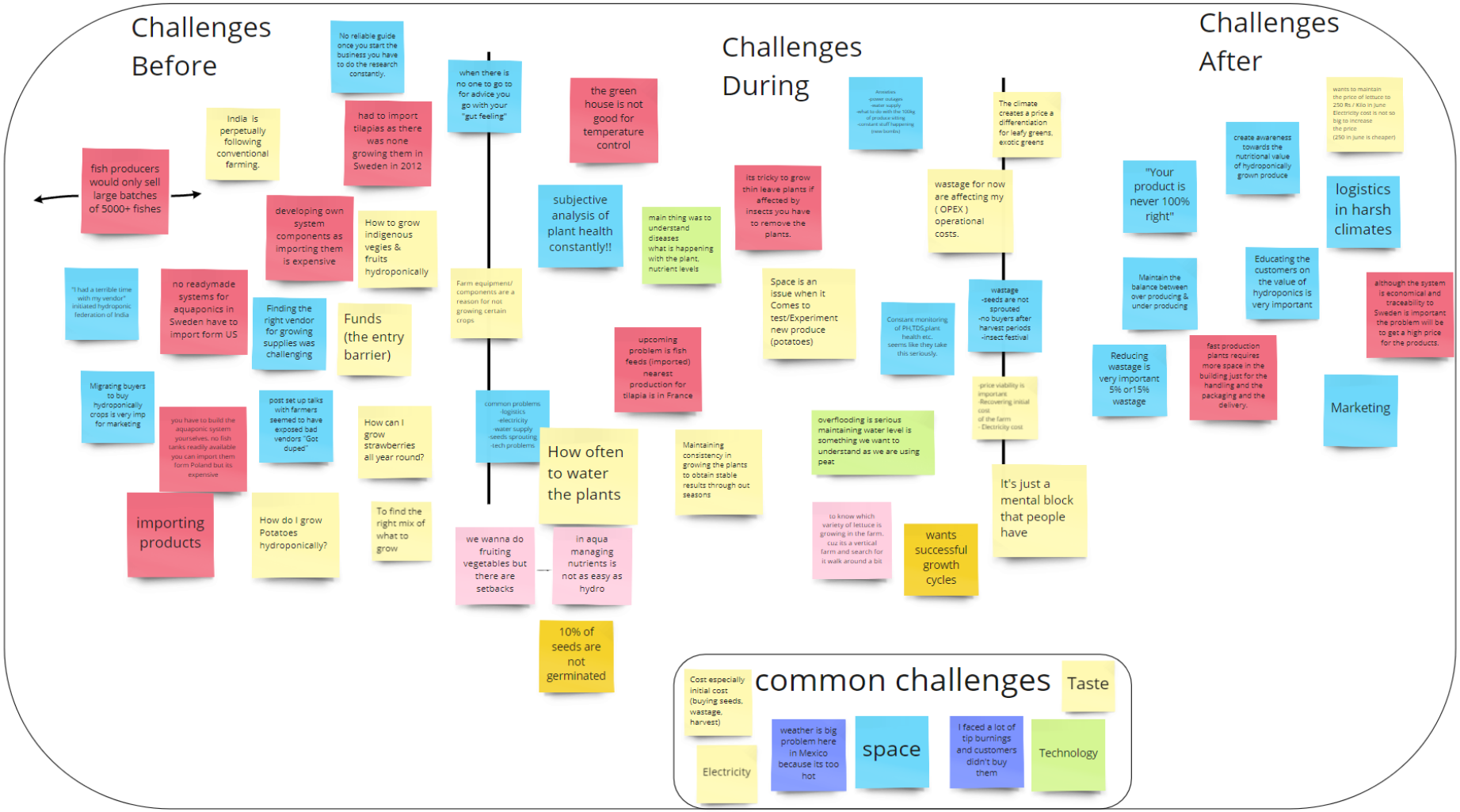

my customers wants traceability
since they have a less shelf life we just want to harvest them and send them to the distributer
To have a produce which is safe, healthy & nutritious
selling to the supermarkets is one goal
people would not hesitate to pay 2x the store amount in sweden.
Traceability of the food is important for customers!
we need to be nearby at the time of the delivery and just get them out of here!
Customers
I produce 100-120 kg of bananas each year
we are growers and also we are providers of the NFT system is as a service
proximity to customer is key
proximity of clients is in the radius of 3 km from the farm.
consumption of staple food is high (India)
I decided to grow lettuce because of the customer demand here in Mexico


plant data
All data helps to minimize the economic risk for my customers
every seed takes 45-50days to grow
The farm help me optimize my vertical tower product.
was not willing to show how many days it takes them to grow
green house is to try out -new species and how to grow them -harvest times
learn the labor/ manhandling required in growing the plants
I'm using NFT but I'm also experimenting with DWC
testing space in the farm for research
what is widely used in India & how can they be grown.
Currently test 3 different types of papayas
R&D
To understand what is widely used in India & how can they be grown
I have 3 students to help me with my experiments

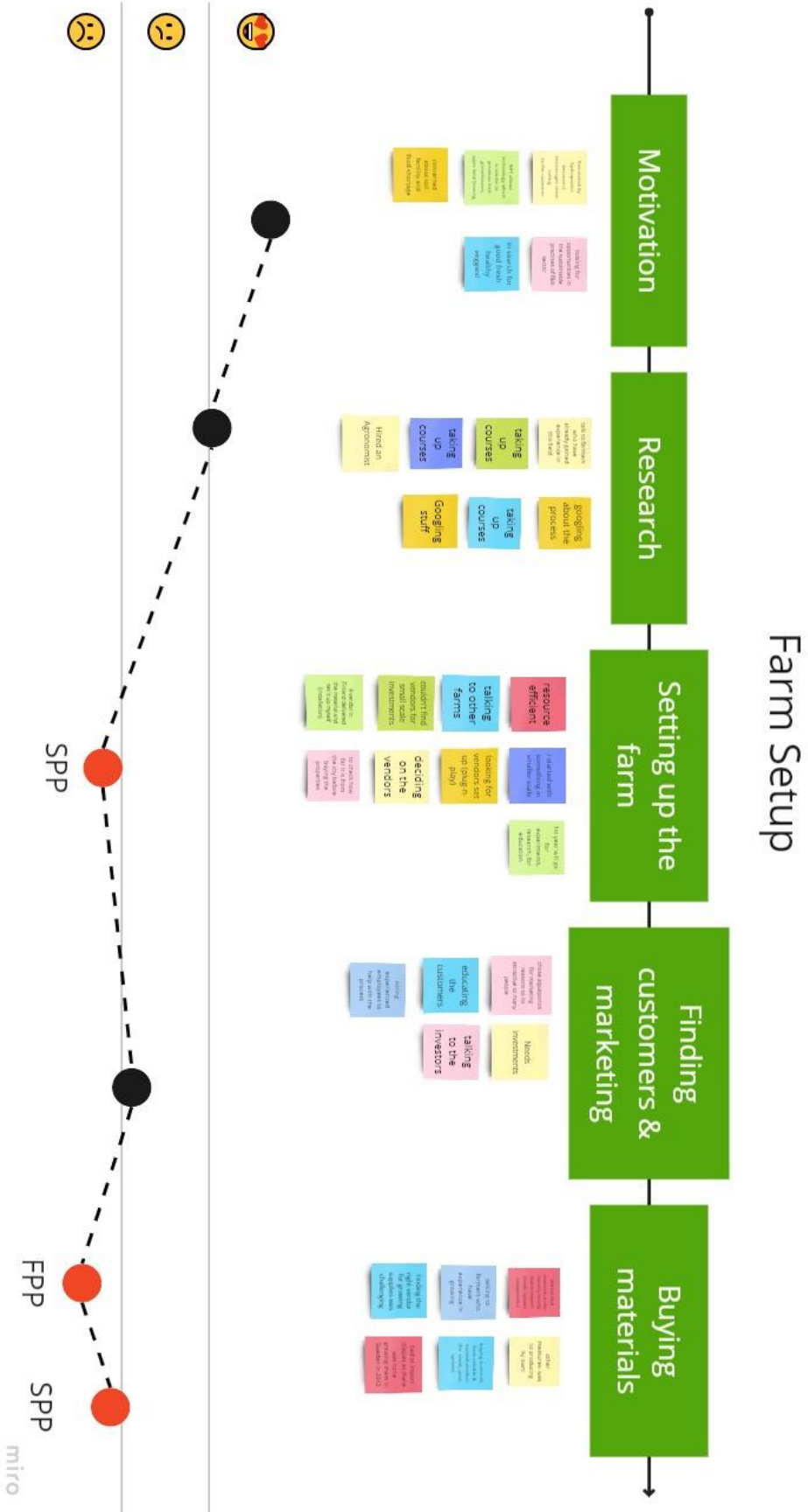

wants to scale up and move to different cities
The main goal is to increase the domestic production in Sweden.
contributing more to sustainable farming
Main aim is to sell individual towers to home growers
"we need more atomization so I can get fired"
investment in tech that leads to less human interference.
Future
-space -marketing R&D getting better & innovating
"we have a solution infront of us, it's about time we start using it at a bigger scale"
seeking help from an authority/organization to help with setups
standardize the concepts that can be put in production and used in future by both small scale and large scale


how much you sell it for
weekends are hectic since harvesting every week
instructed to harvest a certain amount in terms of packets
the produce that you can't sell is wastage for me
harvest- cold storage (10c) for not more than 20hrs
Post harvest
sowing- nursery-tower- harvest depending on demand.
When Growth speed decreases during the cycle it's time to harvest
weekly deliveries to restaurants
harvesting happens depending on the demand

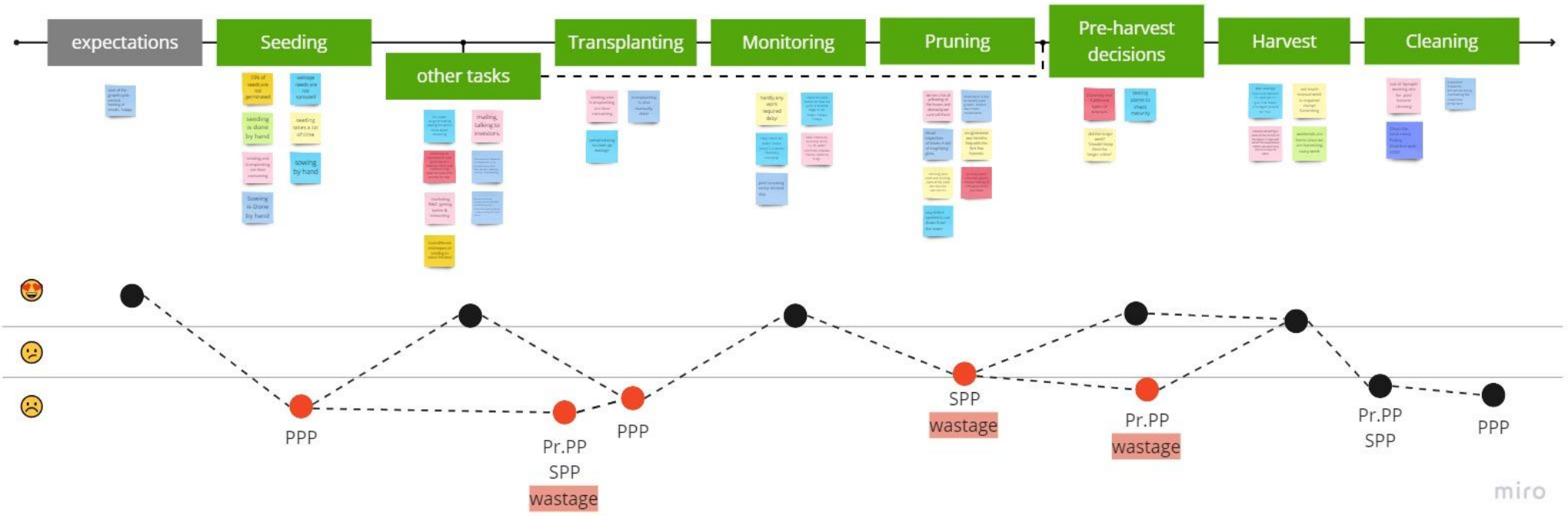

emphasis on investing to preserve plant temperature after harvest
I can use UVA lights to save the models but since I don't have a large greenhouse and also its expensive
constantly looking for new innovations looking to adopt an IOT system
buying seedlings is not an option
I use external products that allows OMRI certification
"everybody wants to be a grower and also a provider of their technology which is contradictory"
Innovation
willing to outsource tech which are in "plug n play" condition
All data helps to minimize the economic risk for my customers
developing inhouse technology in jaipur
trusted name is important in buying tech
if there is tech to provide some help with plant health it would be great


Challenges Before
No reliable guide once you start the business you have to do the research constantly.
India is perpetually following conventional farming.
had to import tilapias as there was none growing them in Sweden in 2012
when there is no one to go to for advice you go with your "gut feeling"
the green house is not good for temperature control
fish producers would only sell large batches of 5000+ fishes
developing own system components as importing them is expensive
How to grow indigenous vegies & fruits hydroponically
subjective analysis of plant health constantly!!
main thing was to understand diseases what is happening with the plant, nutrient levels
its tricky to grow thin leave plants if affected by insects you have to remove the plants.
Challenges During
Annalies -power outages -water supply -what to do with the 100kg of produce sitting -constant stuff happening (new bombs)
The climate creates a price a differentiation for leafy greens, exotic greens
wastage for now are affecting my ( OPEX ) operational costs.
Challenges After
wants to maintain the price of lettuce to 250 Rs / Kilo in June Electricity cost is not so big to increase the price (250 in June is cheaper)
create awareness towards the nutritional value of hydroponically grown produce
"Your product is never 100% right"
logistics in harsh climates
"I had a terrible time with my vendor" initiated hydroponic federation of India
no readymade systems for aquaponics in Sweden have to import form US
Finding the right vendor for growing supplies was challenging
Funds (the entry barrier)
Farm equipment/ components are a reason for not growing certain crops
upcoming problem is fish feeds (imported) nearest production for tilapia is in France
Space is an issue when it Comes to test/Experiment new produce (potatoes)
Constant monitoring of PH,TDS,plant health etc. seems like they take this seriously.
wastage -seeds are not sprouted -no buyers after harvest periods -insect festival
Maintain the balance between over producing & under producing
Educating the customers on the value of hydroponics is very important
although the system is economical and traceability to Sweden is important the problem will be to get a high price for the products.
Migrating buyers to buy hydroponically crops is very imp for marketing
you have to build the aquaponic system yourselves, no fish tanks readily available you can import them form Poland but its expensive
post set up talks with farmers seemed to have exposed bad vendors "Got duped"
How can I grow strawberries all year round?
common problems -logistics -electricity -water supply -seeds sprouting -tech problems
How often to water the plants
Maintaining consistency in growing the plants to obtain stable results through out seasons
overflooding is serious maintaining water level is something we want to understand as we are using peat
-price viability is important -Recovering initial cost of the farm - Electricity cost
Reducing wastage is very important 5% or15% wastage
fast production plants requires more space in the building just for the handling and the packaging and the delivery.
Marketing
importing products
How do I grow Potatoes hydroponically?
To find the right mix of what to grow
we wanna do fruiting vegetables but there are setbacks
in aqua managing nutrients is not as easy as hydro
to know which variety of lettuce is growing in the farm. cuz its a vertical farm and search for it walk around a bit
wants successful growth cycles
It's just a mental block that people have
10% of seeds are not germinated
common challenges
Cost especially initial cost (buying seeds, wastage, harvest)
Taste
weather is big problem here in Mexico because its too hot
space
I faced a lot of tip burnings and customers didn't buy them
Technology
Electricity

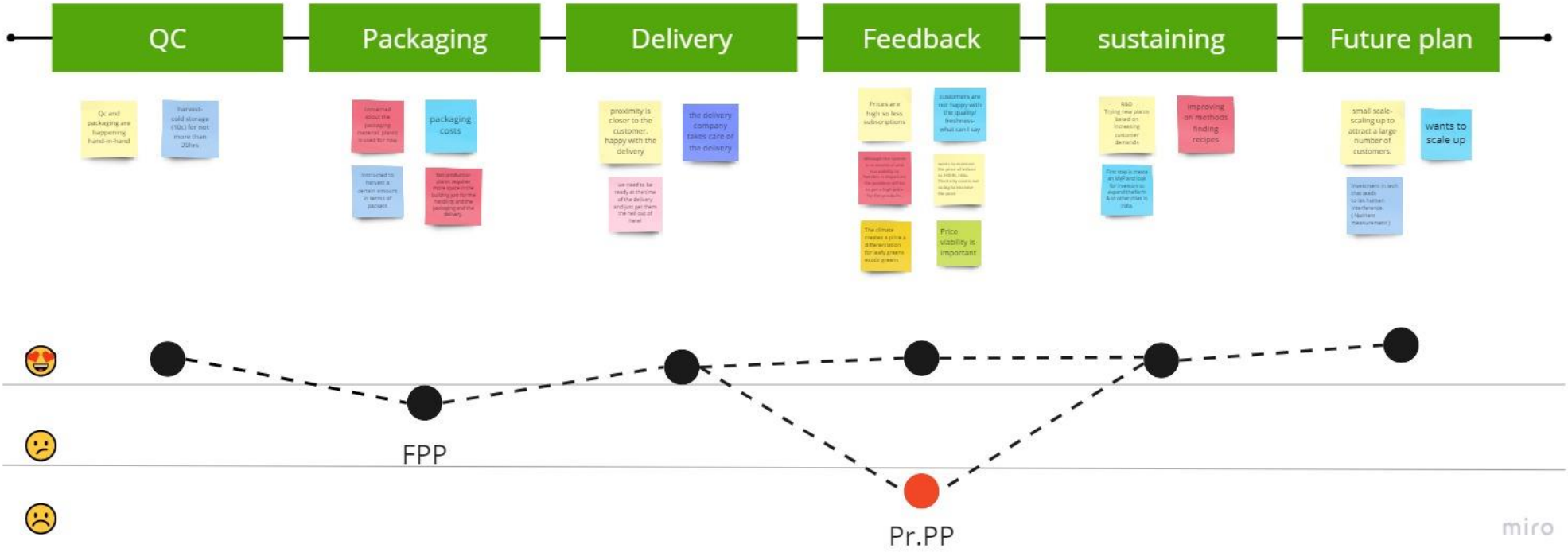
# Appendix E -Journey maps



## Stage 2 -Whole Growing process



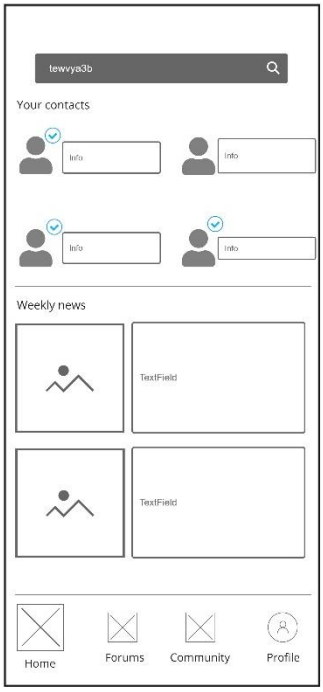
# Post Harvest

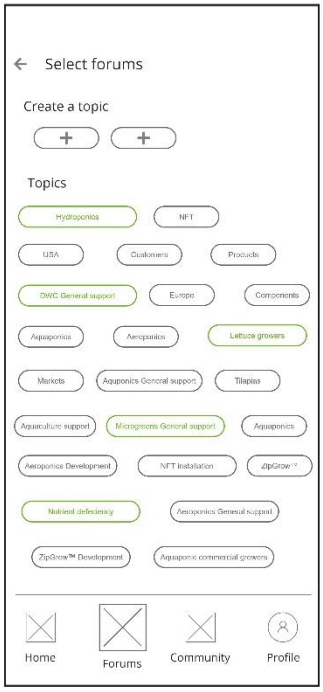


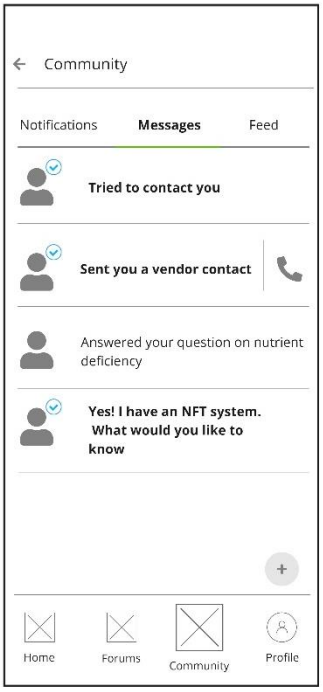
# Appendix F - Survey responses

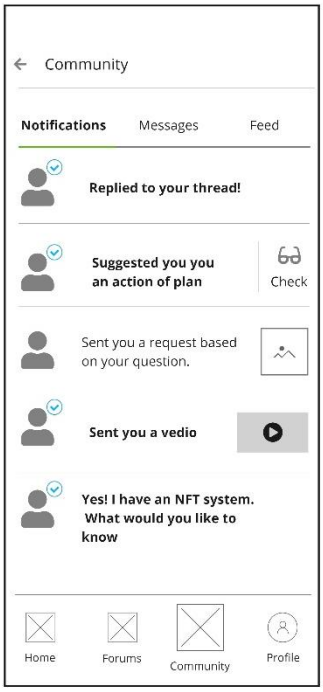
https://docs.google.com/forms/d/1DzgcVMpuxNdwqc_wjhQFScuUVCyixuS2Hk5nyYXLccs/viewanalytics

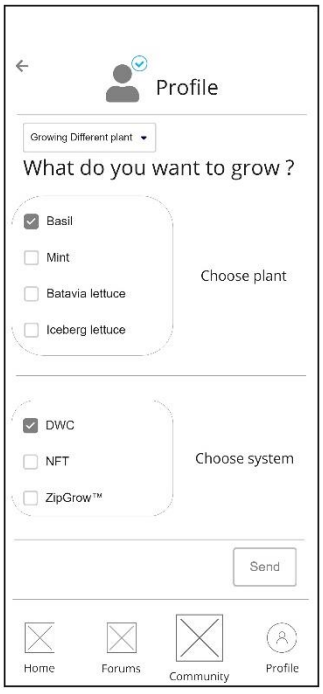
# Appendix G – Low fidelity prototype - Concept 1

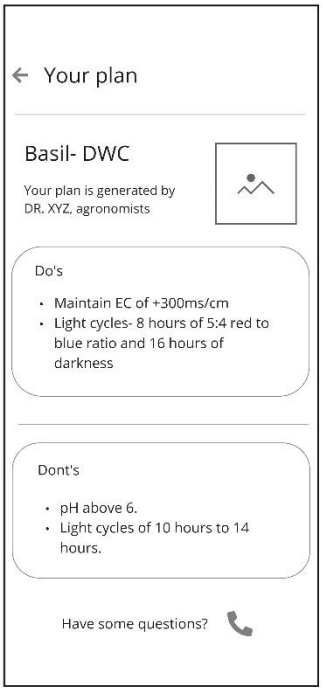


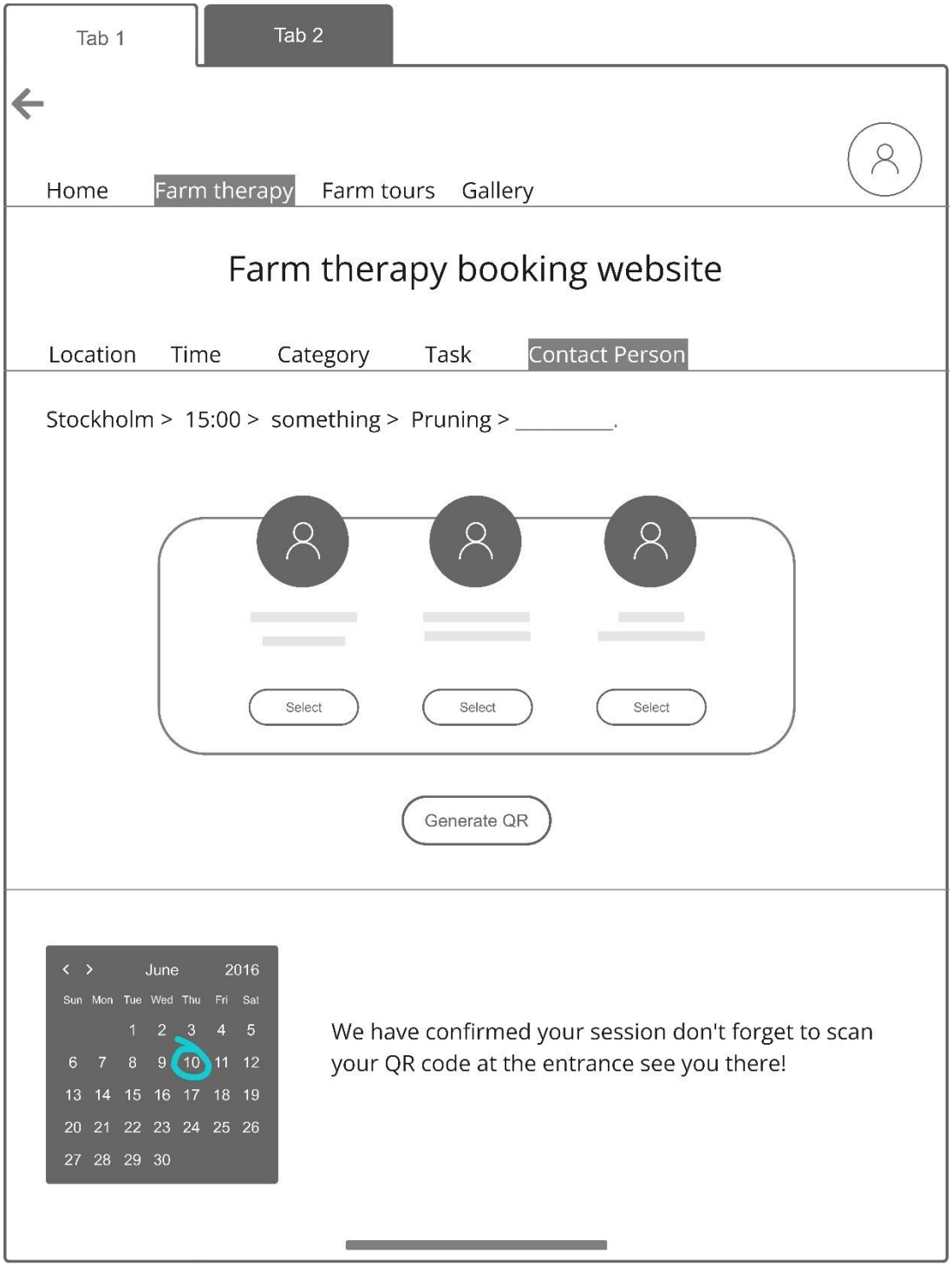



# Concept 2


Tab 1
Tab 2
Home
Farm therapy
Farm tours
Gallery
Farm therapy booking website
Location
Time
Category
Task
Contact Person
Stockholm > 15:00 > something > Pruning > _____.
Select
Select
Select
Generate QR
June
2016
Sun Mon Tue Wed Thu Fri Sat
1 2 3 4 5
6 7 8 9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30
We have confirmed your session don't forget to scan your QR code at the entrance see you there!

**Farm images for assigning task:**





TRITA-ITM-EX 2022:30